

User Manual

for S32K3XX CANTRCV_TJA1145A Driver

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Chapter 1

Revision History

Revision	Date	Author	Description
1.0	May 22 2023	NXP AASW Team	CANTRCV TJA1145A AUTOSAR R21-11 Version 0.8.0

Chapter 2

Introduction

- [Supported Derivatives](#)
- [Overview](#)
- [About This Manual](#)
- [Acronyms and Definitions](#)
- [Reference List](#)

This User Manual describes NXP Semiconductor AUTOSAR CanTrcv driver. AUTOSAR CanTrcv driver configuration parameters and deviations from the specification are described in Driver chapter of this document. AUTOSAR CanTrcv driver requirements and APIs are described in the AUTOSAR CanTrcv driver software specification document.

2.1 Supported Derivatives

The software described in this document is intended to be used with the following microcontroller devices of NXP Semiconductors:

- s32k310_mqfp100
- s32k310_lqfp48
- s32k311_mqfp100 / MWCT2015S_mqfp100
- s32k311_lqfp48
- s32k312_mqfp100 / MWCT2016S_mqfp100
- s32k312_mqfp172 / MWCT2016S_mqfp172
- s32k314_mqfp172
- s32k314_mapbga257
- s32k322_mqfp100 / MWCT2D16S_mqfp100
- s32k322_mqfp172 / MWCT2D16S_mqfp172

- s32k324_mqfp172 / MWCT2D17S_mqfp172
- s32k324_mapbga257
- s32k341_mqfp100
- s32k341_mqfp172
- s32k342_mqfp100
- s32k342_mqfp172
- s32k344_mqfp172
- s32k344_mapbga257
- s32k394_mapbga289
- s32k396_mapbga289
- s32k358_mqfp172
- s32k358_mapbga289
- s32k328_mqfp172
- s32k328_mapbga289
- s32k338_mqfp172
- s32k338_mapbga289
- s32k348_mqfp172
- s32k348_mapbga289
- s32m274_lqfp64
- s32m276_lqfp64

All of the above microcontroller devices are collectively named as S32K3.

Note: MWCT part numbers contain NXP confidential IP for Qi Wireless Power.

2.2 Overview

AUTOSAR (AUTomotive Open System ARchitecture) is an industry partnership working to establish standards for software interfaces and software modules for automobile electronic control systems.

AUTOSAR:

- paves the way for innovative electronic systems that further improve performance, safety and environmental friendliness.
- is a strong global partnership that creates one common standard: "Cooperate on standards, compete on implementation".
- is a key enabling technology to manage the growing electrics/electronics complexity. It aims to be prepared for the upcoming technologies and to improve cost-efficiency without making any compromise with respect to quality.
- facilitates the exchange and update of software and hardware over the service life of the vehicle.

2.3 About This Manual

This Technical Reference employs the following typographical conventions:

- **Boldface** style: Used for important terms, notes and warnings.
- *Italic* style: Used for code snippets in the text. Note that C language modifiers such "const" or "volatile" are sometimes omitted to improve readability of the presented code.

Notes and warnings are shown as below:

Note

This is a note.

Warning

This is a warning

2.4 Acronyms and Definitions

Term	Definition
API	Application Programming Interface
ASM	Assembler
BSMI	Basic Software Make file Interface
CAN	Controller Area Network
C/CPP	C and C++ Source Code
CS	Chip Select
CTU	Cross Trigger Unit
DEM	Diagnostic Event Manager
DET	Development Error Tracer
DMA	Direct Memory Access
ECU	Electronic Control Unit
FIFO	First In First Out
LSB	Least Significant Bit
MCU	Micro Controller Unit
MIDE	Multi Integrated Development Environment
MSB	Most Significant Bit
N/A	Not Applicable
RAM	Random Access Memory
SIU	Systems Integration Unit
SWS	Software Specification
VLE	Variable Length Encoding
XML	Extensible Markup Language

2.5 Reference List

#	Title	Version
1	Specification of CanTrcv Driver	AUTOSAR Release R21-11
2	TJA1145A objective datasheet	2

Chapter 3

Driver

- [Requirements](#)
- [Driver Design Summary](#)
- [Hardware Resources](#)
- [Deviations from Requirements](#)
- [Driver Limitations](#)
- [Driver usage and configuration tips](#)
- [Runtime errors](#)
- [Symbolic Names Disclaimer](#)

3.1 Requirements

Requirements for this driver are detailed in the AUTOSAR R21-11 (4.7) CanTrcv Driver Software Specification document.

3.2 Driver Design Summary

The CanTrcv driver configures and controls the TJA1145A transceiver.

3.3 Hardware Resources

The hardware configured by the CanTrcv driver is the TJA1145A transceiver.

3.4 Deviations from Requirements

The driver deviates from the AUTOSAR CanTrcv Driver software specification in some places. The table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the CanTrcv Driver.

Term	Definition
N/A	Not available
N/T	Not testable
N/S	Out of scope
N/I	Not implemented
N/F	Not fully implemented

Below table identifies the AUTOSAR requirements that are not fully implemented, implemented differently, not available, not testable or out of scope for the driver.

Requirement	Status	Description
ECUC_CanTrcv_00097	N/A	TJA1145 does not control ECU power supply.
ECUC_CanTrcv_00145	N/A	TJA1145 is configurable via SPI.
ECUC_CanTrcv_00149	N/A	TJA1145 is configurable via SPI.
ECUC_CanTrcv_00150	N/A	TJA1145 is configurable via SPI.
ECUC_CanTrcv_00157	N/A	TJA1145 is configurable via SPI.
ECUC_CanTrcv_00185	N/A	TJA1145A doesn't require usage of input capture unit (ICU) because the only interrupt source is RxD pin which is handled by CAN driver.
ECUC_CanTrcv_00190	N/A	Delays using tm.
ECUC_CanTrcv_00191	N/A	Delays using tm.
SWS_CanTrcv_00171	N/A	TJA1145A doesn't require usage of input capture unit (ICU) because the only interrupt source is RxD pin which is handled by CAN driver.
SWS_CanTrcv_00172	N/A	TJA1145A doesn't require usage of input capture unit (ICU) because the only interrupt source is RxD pin which is handled by CAN driver.
SWS_CanTrcv_00173	N/A	TJA1145A doesn't require usage of input capture unit (ICU) because the only interrupt source is RxD pin which is handled by CAN driver.
SWS_CanTrcv_00182	N/I	Driver offers POR and SYSERR monitoring API.
SWS_CanTrcv_00230	N/A	The driver doesn't use timers for state change.
SWS_CanTrcv_00999	N/A	Listing no longer valid requirements in this specification.

3.5 Driver Limitations

3.6 Driver usage and configuration tips

None.

3.7 Runtime errors

The driver generates the following errors at runtime.

- Default Errors (reported by DET):

Function	Error Code	Condition triggering the error
Most functions	CANTRCV_43_TJA1145A_E_↔ UNINIT	driver uninitialized
Most functions	CANTRCV_43_TJA1145A_E_↔ INVALID_TRANSCEIVER	parameter out of range
Most functions	CANTRCV_43_TJA1145A_E_↔ PARAM_POINTER	NULL_PTR
Most functions	CANTRCV_43_TJA1145A_E_↔ NO_TRCV_CONTROL	error in SPI communication
CanTrcv_43_tja1145a_Init	CANTRCV_43_TJA1145A_E_↔ INIT_FAILED	driver already initialized, number of devices is 0 or higher than maximum allowed
CanTrcv_43_tja1145a_Init	CANTRCV_43_TJA1145A↔ _E_BAUDRATE_NOT_↔ SUPPORTED	requested baud rate not supported
CanTrcv_43_tja1145a_SetOp↔ Mode	CANTRCV_43_TJA1145A_E_↔ PARAM_TRCV_OPMODE	invalid operating mode
CanTrcv_43_tja1145a_SetOp↔ Mode	CANTRCV_43_TJA1145A_E_↔ TRCV_NOT_NORMAL	invalid mode transition (sleep to standby)
CanTrcv_43_tja1145a_SetOp↔ Mode	CANTRCV_43_TJA1145A_E_↔ TRCV_NOT_STANDBY	invalid mode transition (normal to sleep)
CanTrcv_43_tja1145a_Set↔ WakeupMode	CANTRCV_43_TJA1145A_E↔ _PARAM_TRCV_WAKEUP_↔ MODE	requested mode not supported
CanTrcv_43_tja1145a_DeInit	CANTRCV_43_TJA1145A_E_↔ TRCV_NOT_NORMAL	invalid mode

3.8 Symbolic Names Disclaimer

All containers having symbolicNameValue set to TRUE in the AUTOSAR schema will generate defines like:

```
#define <Mip>Conf_<Container_ShortName>_<Container_ID>
```

For this reason it is forbidden to duplicate the names of such containers across the RTD configurations or to use names that may trigger other compile issues (e.g. match existing #ifdefs arguments).

Chapter 4

Tresos Configuration Plug-in

This chapter describes the Tresos configuration plug-in for the driver. All the parameters are described below.

- Module [CanTrcv](#)
 - Container [CanTrcvGeneral](#)
 - * Parameter [CanTrcvWakeUpSupport](#)
 - * Parameter [CanTrcvMainFunctionDiagnosticsPeriod](#)
 - * Parameter [CanTrcvMainFunctionPeriod](#)
 - * Parameter [CanTrcvDevErrorDetect](#)
 - * Parameter [CanTrcvDiagnosticEventManager](#)
 - * Parameter [CanTrcvEnableUserModeSupport](#)
 - * Parameter [CanTrcvMulticoreSupport](#)
 - * Parameter [CanTrcvTimerType](#)
 - * Parameter [CanTrcvWaitTime](#)
 - * Parameter [CanTrcvVersionInfoApi](#)
 - * Parameter [CanTrcvIndex](#)
 - * Reference [CanTrcvEcucPartitionRef](#)
 - Container [CanTrcvConfigSet](#)
 - * Parameter [CanTrcvSPICommRetries](#)
 - * Parameter [CanTrcvSPICommTimeout](#)
 - * Parameter [CanTrcvSPILevelDelivered](#)
 - * Parameter [CanTrcvSpiAccessSynchronous](#)
 - * Container [CanTrcvChannel](#)
 - Parameter [CanTrcvChannelId](#)
 - Parameter [CanTrcvInitState](#)
 - Parameter [CanTrcvMaxBaudrate](#)
 - Parameter [CanTrcvChannelUsed](#)
 - Parameter [CanTrcvControlsPowerSupply](#)
 - Parameter [CanTrcvWakeupByBusUsed](#)
 - Parameter [CanTrcvHwPnSupport](#)
 - Reference [CanTrcvIcuChannelRef](#)
 - Reference [CanTrcvPorWakeupSourceRef](#)
 - Reference [CanTrcvSyserrWakeupSourceRef](#)

- Reference [CanTrcvWakeupSourceRef](#)
- Reference [CanTrcvChannelEcucPartitionRef](#)
- Container [CanTrcvAccess](#)
- Container [CanTrcvDemEventParameterRefs](#)
- Reference [CANTRCV_E_BUS_ERROR](#)
- Container [CanTrcvPartialNetwork](#)
- Parameter [CanTrcvBaudRate](#)
- Parameter [CanTrcvPnFrameCanId](#)
- Parameter [CanTrcvPnFrameCanIdMask](#)
- Parameter [CanTrcvPnFrameDlc](#)
- Parameter [CanTrcvFdEnable](#)
- Parameter [CanTrcvPnCanIdIsExtended](#)
- Parameter [CanTrcvPnEnabled](#)
- Parameter [CanTrcvBusErrFlag](#)
- Parameter [CanTrcvPowerOnFlag](#)
- Parameter [WufPattern](#)
- Container [CanTrcvPnFrameDataMaskSpec](#)
- Parameter [CanTrcvPnFrameDataMask](#)
- Parameter [CanTrcvPnFrameDataMask1](#)
- Parameter [CanTrcvPnFrameDataMask2](#)
- Parameter [CanTrcvPnFrameDataMask3](#)
- Parameter [CanTrcvPnFrameDataMask4](#)
- Parameter [CanTrcvPnFrameDataMask5](#)
- Parameter [CanTrcvPnFrameDataMask6](#)
- Parameter [CanTrcvPnFrameDataMask7](#)
- Parameter [CanTrcvPnFrameDataMaskIndex](#)
- Container [WuConfig](#)
- Parameter [CanBusWuConfig](#)
- Parameter [WakePinConfig](#)
- Container [EventCaptureConfig](#)
- Parameter [OvertempDetection](#)
- Parameter [SpiFailureDetection](#)
- Parameter [CanBusSilenceDetection](#)
- Parameter [CanBusFailureDetection](#)
- Parameter [vccUvDetection](#)
- Container [LockControl](#)
- Parameter [LockRange06h09h](#)
- Parameter [LockRange10h1Fh](#)
- Parameter [LockRange20h2Fh](#)
- Parameter [LockRange30h3Fh](#)
- Parameter [LockRange40h4Fh](#)
- Parameter [LockRange50h5Fh](#)
- Parameter [LockRange68h6Fh](#)
- Container [CommonPublishedInformation](#)
 - * Parameter [ArReleaseMajorVersion](#)
 - * Parameter [ArReleaseMinorVersion](#)

- * Parameter [ArReleaseRevisionVersion](#)
- * Parameter [ModuleId](#)
- * Parameter [SwMajorVersion](#)
- * Parameter [SwMinorVersion](#)
- * Parameter [SwPatchVersion](#)
- * Parameter [VendorApiInfix](#)
- * Parameter [VendorId](#)

4.1 Module CanTrcv

This container holds the configuration of a single CanTrcv Driver.

Included containers:

- [CanTrcvGeneral](#)
- [CanTrcvConfigSet](#)
- [CommonPublishedInformation](#)

Property	Value
type	ECUC-MODULE-DEF
lowerMultiplicity	1
upperMultiplicity	Infinite
postBuildVariantSupport	true
supportedConfigVariants	VARIANT-LINK-TIME, VARIANT-POST-BUILD, VARIANT-PRE-COMPILE

4.2 Container CanTrcvGeneral

This container holds parameters related to each CanTrcv Driver.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.3 Parameter CanTrcvWakeUpSupport

CANTRCV_WAKEUP_BY_POLLING: Wake up by polling.

CANTRCV_WAKEUP_NOT_SUPPORTED: Wake up not supported.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	CANTRCV_WAKEUP_BY_POLLING
literals	['CANTRCV_WAKEUP_BY_POLLING', 'CANTRCV_WAKEUP_NOT_SUPPORTED']

4.4 Parameter CanTrcvMainFunctionDiagnosticsPeriod

This parameter describes the period for cyclic call to CanTrcv_MainFunctionDiagnostics.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	255.0
min	0.0

4.5 Parameter CanTrcvMainFunctionPeriod

This parameter describes the period for cyclic call to CanTrcv_MainFunction.

Property	Value
type	ECUC-FLOAT-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	0.0
max	255.0
min	0.0

4.6 Parameter CanTrcvDevErrorDetect

Switches the Development Error Detection and Notification: ON or OFF.

When this option is OFF code size is reduced, but no error detection is available.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.7 Parameter CanTrcvDiagnosticEventManager

Enable or Disable the API for reporting the Dem Error.

Note: Implementation Specific Parameter.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.8 Parameter CanTrcvEnableUserModeSupport

When this parameter is enabled, the CanTrcv module will adapt to run from User Mode.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.9 Parameter CanTrcvMulticoreSupport

Enable Maps CanTrcv driver to multiple EcuC partitions to make the modules API available in this partition. The CanTrcv driver will operate as an independent instance in each of the partitions.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.10 Parameter CanTrcvTimerType

Type of the Time Service Predefined Timer.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	None
literals	['None', 'Timer_1us16bit']

4.11 Parameter CanTrcvWaitTime

Wait time for transceiver state changes in seconds.

Property	Value
type	ECUC-FLOAT-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0.0
max	2.55E-4
min	0.0

4.12 Parameter CanTrcvVersionInfoApi

Switches the CanTrcv_GetVersionInfo() API: ON or OFF.

When this option is ON, the driver supports API for getting Version information of the Driver.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.13 Parameter CanTrcvIndex

Specifies the InstanceId of this module instance.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	15
min	0

4.14 Reference CanTrcvEcucPartitionRef

Maps the CAN transceiver driver to zero or multiple ECUC partitions to make the modules API available in this partition.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	Infinite
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/EcuC/EcucPartitionCollection/EcucPartition

4.15 Container CanTrcvConfigSet

This is the multiple configuration set container for CanTrcv Driver.

Included subcontainers:

- [CanTrcvChannel](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.16 Parameter CanTrcvSPICommRetries

Indicates the maximum number of communication retries in case of a failed SPI communication (applies both to timed out communication and to errors/NACK in the response data).

If configured value is '0', no retry is allowed (communication is expected to succeed at first try).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.17 Parameter CanTrcvSPICommTimeout

Indicates the maximum time allowed to the CanTrcv to reply (either positively or negatively) to a SPI command.

Timeout is configured in milliseconds. Timeout value of '0' means no specific timeout is to be used by CanTrcv and the communication is executed at the best of the SPI HW capacity.

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	AUTOSAR_ECUC
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	100
min	0

4.18 Parameter CanTrcvSPILevelDelivered

Selects transmission behaviour. This value must match SpiLevelDelivered in SPI configuration.

Level 0 - simple synchronous behavior

Level 1 - basic asynchronous behaviour

Level 2 - enhanced behaviour

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	False
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	2
min	0

4.19 Parameter CanTrcvSpiAccessSynchronous

Defines whether the access to the Spi sequence is synchronous or asynchronous.

true: SPI access is synchronous.

false: SPI access is asynchronous.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.20 Container CanTrcvChannel

Defines a structure to initialize the selected CAN transceiver (channel).

Included subcontainers:

- [CanTrcvAccess](#)
- [CanTrcvDemEventParameterRefs](#)
- [CanTrcvPartialNetwork](#)
- [WuConfig](#)
- [EventCaptureConfig](#)
- [LockControl](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	16
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE

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4.21 Parameter CanTrcvChannelId

Unique identifier of the CAN Transceiver Channel.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	15
min	0

4.22 Parameter CanTrcvInitState

State of CAN transceiver after call to CanTrcv_Init.

- Normal mode after Init
- Sleep mode after Init
- Standby mode after Init (Default)

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	CANTRCV_OP_MODE_NORMAL
literals	['CANTRCV_OP_MODE_NORMAL', 'CANTRCV_OP_MODE_SLEEP', 'CANTRCV_OP_MODE_STANDBY']

4.23 Parameter CanTrcvMaxBaudrate

Indicates the data transfer rate in kbps. Maximum data transfer rate in kbps for transceiver hardware type.

For CAN the maximum data transfer rate is 500 kbps, for CAN FD 2000 kbps.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	1000
max	12000
min	0

4.24 Parameter CanTrcvChannelUsed

Shall the related CAN transceiver be used?

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.25 Parameter CanTrcvControlsPowerSupply

Is ECU power supply controlled by this transceiver?

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.26 Parameter CanTrcvWakeupByBusUsed

Is wake up by bus supported?

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	true

4.27 Parameter CanTrcvHwPnSupport

Indicates whether the HW supports the selective wake-up functionality.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.28 Reference CanTrcvIcuChannelRef

Reference to the IcuChannel to enable/disable the interrupts for wakeups.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcucDefs/Icu/IcuConfigSet/IcuChannel

4.29 Reference CanTrcvPorWakeupSourceRef

Symbolic name reference to specify the wakeup sources that should be used in the calls to EcuM_SetWakeupEvent.

Property	Value
type	ECUC-REFERENCE-DEF

Property	Value
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuM/EcuMConfiguration/EcuMCommon↔ Configuration/EcuMWakeupSource

4.30 Reference CanTrcvSyserrWakeupSourceRef

Symbolic name reference to specify the wakeup sources that should be used in the calls to EcuM_SetWakeupEvent.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuM/EcuMConfiguration/EcuMCommon↔ Configuration/EcuMWakeupSource

4.31 Reference CanTrcvWakeupSourceRef

Reference to a wakeup source in the EcuM configuration.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuM/EcuMConfiguration/EcuMCommon← Configuration/EcuMWakeupSource

4.32 Reference CanTrcvChannelEcucPartitionRef

Maps the CAN transceiver configuration to zero or one ECUC partitions.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	true
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/EcuC/EcucPartitionCollection/EcucPartition

4.33 Container CanTrcvAccess

Gives CanTrcv Driver information about access to a single CAN transceiver.

Included choices:

- CanTrcvSpiAccess
- CanTrcvDioAccess

Property	Value
type	ECUC-CHOICE-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.34 Container CanTrcvDemEventParameterRefs

Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameter's DemEventId symbolic value. The standardized errors are provided in this container and can be extended by vendor-specific error references.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.35 Reference CANTRCV_E_BUS_ERROR

Reference to the DemEventParameter which shall be issued when bus error has occurred.

Property	Value
type	ECUC-REFERENCE-DEF
origin	AUTOSAR_ECUC
lowerMultiplicity	0
upperMultiplicity	1
postBuildVariantMultiplicity	false
multiplicityConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PRE-COMPILE
	VARIANT-POST-BUILD: PRE-COMPILE
	VARIANT-PRE-COMPILE: PRE-COMPILE
requiresSymbolicNameValue	False
destination	/AUTOSAR/EcuDefs/Dem/DemConfigSet/DemEventParameter

4.36 Container CanTrcvPartialNetwork

Container gives CAN transceiver driver information about the configuration of Partial Networking functionality.

Included subcontainers:

- [CanTrcvPnFrameDataMaskSpec](#)

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.37 Parameter CanTrcvBaudRate

Indicates the data transfer rate in kbps.

Supported values:

- 50 kBits
- 100 kBits
- 125 kBits
- 250 kBits
- 500 kBits
- 1000 kBits

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	50
max	12000
min	0

4.38 Parameter CanTrcvPnFrameCanId

CAN ID of the Wake-up Frame (WUF).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	4294967295
min	0

4.39 Parameter CanTrcvPnFrameCanIdMask

ID Mask for the selective activation of the transceiver. It is used to enable-Frame Wake-up (WUF) on a group of IDs.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	4294967295
min	0

4.40 Parameter CanTrcvPnFrameDlc

Data Length of the Wake-up Frame (WUF).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	1
max	8
min	0

4.41 Parameter CanTrcvFdEnable

Enable/disable flexible data rate.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	TJA1145A_FD_DISABLED
literals	['TJA1145A_FD_ENABLED', 'TJA1145A_FD_DISABLED']

4.42 Parameter CanTrcvPnCanIdIsExtended

Indicates whether extended or standard ID is used.

TRUE = Extended Can identifier is used.

FALSE = Standard Can identifier is used

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.43 Parameter CanTrcvPnEnabled

Indicates whether the selective wake-up function is enabled or disabled in HW.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.44 Parameter CanTrcvBusErrFlag

Indicates if the Bus Error (BUSERR) flag is managed by the BSW. This flag is set if a bus failure is detected by the transceiver.

TRUE = Supported by transceiver and managed by BSW. FALSE = Not managed by BSW.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.45 Parameter CanTrcvPowerOnFlag

Description: Indicates if the Power On Reset (POR) flag is available and is managed by the transceiver.

TRUE = Supported by Hardware.

FALSE = Not supported by Hardware

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	true

4.46 Parameter WufPattern

Determines whether DLC and data payload will be included in wake-up frame pattern.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	TJA1145A_WUF_ID
literals	['TJA1145A_WUF_ID', 'TJA1145A_WUF_ID_DLC_DATA']

4.47 Container CanTrcvPnFrameDataMaskSpec

Defines data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received wake-up frame (WUF).

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.48 Parameter CanTrcvPnFrameDataMask

Defines the byte 0 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.49 Parameter CanTrcvPnFrameDataMask1

Defines the byte 1 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A

Property	Value
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.50 Parameter CanTrcvPnFrameDataMask2

Defines the byte 2 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.51 Parameter CanTrcvPnFrameDataMask3

Defines the byte 3 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.52 Parameter CanTrcvPnFrameDataMask4

Defines the byte 4 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.53 Parameter CanTrcvPnFrameDataMask5

Defines the byte 5 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP

Property	Value
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.54 Parameter CanTrcvPnFrameDataMask6

Defines the byte 6 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF)..

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.55 Parameter CanTrcvPnFrameDataMask7

Defines the byte 7 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	255
min	0

4.56 Parameter CanTrcvPnFrameDataMaskIndex

Holds the position n in frame of the mask-part.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	AUTOSAR_ECUC
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	0
max	7
min	0

4.57 Container WuConfig

Wake up configuration.

Included subcontainers:

Tresos Configuration Plug-in

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.58 Parameter CanBusWuConfig

Enables/disables wake up by CAN bus.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	TJA1145A_WU_CAN_BUS_ENABLED
literals	['TJA1145A_WU_CAN_BUS_ENABLED', 'TJA1145A_WU_CAN_BUS_↔DISABLED']

4.59 Parameter WakePinConfig

Enable/disable WAKE pin wake up. TJA1145A can be woken up by rising or falling edge of WAKE pin.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true

Property	Value
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	TJA1145A_WU_PIN_RISING_EDGE
literals	['TJA1145A_WU_PIN_RISING_EDGE', 'TJA1145A_WU_PIN_FALLING←_EDGE', 'TJA1145A_WU_PIN_DISABLED']

4.60 Container EventCaptureConfig

Configuration of system/bus event.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.61 Parameter OvertempDetection

Enable/disable overtemperature detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.62 Parameter SpiFailureDetection

Enable/disable SPI failure detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.63 Parameter CanBusSilenceDetection

Enable/disable CAN bus silence detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.64 Parameter CanBusFailureDetection

Enable/disable CAN bus failure detection.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.65 Parameter vccUvDetection

VCC undervoltage detection in active mode configuration.

Property	Value
type	ECUC-ENUMERATION-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	TJA1145A_VCC_UV_DISABLED
literals	['TJA1145A_VCC_UV_DISABLED', 'TJA1145A_VCC_UV_ENABLED']

4.66 Container LockControl

Locks memory areas, which forbids SPI write access.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.67 Parameter LockRange06h09h

Locks register address range from 06h to 09h.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
default Value	false

4.68 Parameter LockRange10h1Fh

Locks register address range from 10h to 1Fh.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD

Property	Value
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.69 Parameter LockRange20h2Fh

Locks register address range from 20h to 2Fh.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.70 Parameter LockRange30h3Fh

Locks register address range from 30h to 3Fh.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.71 Parameter LockRange40h4Fh

Locks register address range from 40h to 4Fh.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.72 Parameter LockRange50h5Fh

Locks register address range from 50h to 5Fh.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.73 Parameter LockRange68h6Fh

Locks register address range from 68h to 6Fh.

Property	Value
type	ECUC-BOOLEAN-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	true
valueConfigClasses	VARIANT-LINK-TIME: LINK
	VARIANT-POST-BUILD: POST-BUILD
	VARIANT-PRE-COMPILE: PRE-COMPILE
defaultValue	false

4.74 Container CommonPublishedInformation

Common container, aggregated by all modules.

It contains published information about vendor and versions.

Included subcontainers:

- None

Property	Value
type	ECUC-PARAM-CONF-CONTAINER-DEF
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A

4.75 Parameter ArReleaseMajorVersion

Major version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	4
max	4
min	4

4.76 Parameter ArReleaseMinorVersion

Minor version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	7
max	7
min	7

4.77 Parameter ArReleaseRevisionVersion

Revision version number of AUTOSAR specification on which the appropriate implementation is based on.

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1

Property	Value
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.78 Parameter ModuleId

Module ID of this module from Module List.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	70
max	70
min	70

4.79 Parameter SwMajorVersion

Major version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF

Property	Value
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.80 Parameter SwMinorVersion

Minor version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	8
max	8
min	8

4.81 Parameter SwPatchVersion

Patch level version number of the vendor specific implementation of the module. The numbering is vendor specific.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	0
max	0
min	0

4.82 Parameter VendorApiInfix

In driver modules which can be instantiated several times on a single ECU, BSW00347 requires that the name of APIs is extended by the VendorId and a vendor specific name.

This parameter is used to specify the vendor specific name. In total, the Implementation specific name is generated as follows:

<ModuleName>__>VendorId>__<VendorApiInfix>.

E.g. assuming that the VendorId of the implementor is 123 and the implementer chose a VendorApiInfix of "v11r456" a api name

Can_Write defined in the SWS will translate to Can_123_v11r456Write.

This parameter is mandatory for all modules with upper multiplicity >

1. It shall not be used for modules with upper multiplicity =1.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-STRING-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION

4.83 Parameter VendorId

Vendor ID of the dedicated implementation of this module according to the AUTOSAR vendor list.

Note: Implementation Specific Parameter

Property	Value
type	ECUC-INTEGER-PARAM-DEF
origin	NXP
symbolicNameValue	false
lowerMultiplicity	1
upperMultiplicity	1
postBuildVariantMultiplicity	N/A
multiplicityConfigClasses	N/A
postBuildVariantValue	false
valueConfigClasses	VARIANT-LINK-TIME: PUBLISHED-INFORMATION
	VARIANT-POST-BUILD: PUBLISHED-INFORMATION
	VARIANT-PRE-COMPILE: PUBLISHED-INFORMATION
defaultValue	43
max	43
min	43

This chapter describes the Tresos configuration plug-in for the CanTrcv Driver. The most of the parameters are described below.

CanTrcv General settings:

CanTrcvWakeUpSupport - Informs whether wake up is supported by polling or not supported.

CanTrcvMainFunctionDiagnosticsPeriod - Sets period for cyclic call to CanTrcv_MainFunctionDiagnostics.

CanTrcvMainFunctionPeriod - Sets period for cyclic call to CanTrcv_MainFunction.

CanTrcvDevErrorDetect - Switches the development error detection and notification on or off. TRUE: detection and notification is enabled. FALSE: detection and notification is disabled.

CanTrcvDiagnosticEventManager - Enables/disables the API for Dem Error reporting.

CanTrcvEnableUserModeSupport - If enabled, Can transceiver module will adapt to run in User Mode.

CanTrcvMulticoreSupport - Maps Can transceiver driver to multiple EcuC partitions to make the modules API available in this partition. The CanTrcv driver will operate as an independent instance in each of the partitions.

CanTrcvTimerType - Type of the Time Service Predefined Timer.

CanTrcvWaitTime - Wait time for transceiver state changes in seconds.

CanTrcvVersionInfoApi - Switches version information API on and off. If switched off, function need not be present in compiled code.

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CanTrcvIndex - Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

CanTrcvEcucPartitionRef - Maps the CAN transceiver driver to zero or multiple ECUC partitions to make the modules API available in this partition. The module will operate as an independent instance in each of the partitions.

CanTrcv Config Set:

CanTrcvSPICommRetries - Indicates the maximum number of communication retries in case of a failed SPI communication (applies both to timed out communication and to errors/NACK in the response data).

CanTrcvSPICommTimeout - Indicates the maximum time allowed for the CanTrcv to reply (either positively or negatively) to a SPI command.

CanTrcvSPILevelDelivered - Selects transmission behaviour. This value must match SpiLevelDelivered in SPI configuration.

CanTrcvSpiAccessSynchronous - Defines whether the access to the Spi sequence is synchronous or asynchronous.

CanTrcv Channel settings:

CanTrcvInitState - State of CAN transceiver after call to CanTrcv_Init.

CanTrcvIcuChannelRef - Reference to the IcuChannel to enable/disable the interrupts for wakeups.

CanTrcvPorWakeupSourceRef - Symbolic name reference to specify the wakeup sources that should be used in the calls to EcuM_SetWakeupEvent.

CanTrcvSyserrWakeupSourceRef - Symbolic name reference to specify the wakeup sources that should be used in the calls to EcuM_SetWakeupEvent.

CanTrcvWakeupSourceRef - Reference to a wakeup source in the EcuM configuration.

CanTrcvChannelId - Unique identifier of the CAN Transceiver Channel.

CanTrcvChannelEcucPartitionRef - Maps the CAN transceiver channel to zero or one ECUC partitions. The ECUC partition referenced is a subset of the ECUC partitions where the CAN transceiver driver is mapped to.

CanTrcvMaxBaudrate - Indicates the data transfer rate in kbps.

CanTrcvChannelUsed - Shall the related CAN transceiver channel be used?

CanTrcvControlsPowerSupply - Is ECU power supply controlled by this transceiver?

CanTrcvWakeupByBusUsed - Is wake up by bus supported?

CanTrcvHwPnSupport - Indicates whether the HW supports the selective wake-up functionality.

CanTrcvSpiAccess:

CanTrcvSpiSequence - Container gives CAN transceiver driver information about one SPI sequence. One SPI sequence used by CAN transceiver driver is in exclusive use for it. No other driver is allowed to access this sequence. CAN transceiver driver may use one sequence to access n CAN transceiver hardware chips of the same type or n sequences are used to access one single CAN transceiver hardware chip. If a CAN transceiver hardware has no SPI interface, there is no instance of this container.

SpiChannelRef - Identifies SPI communication channel used by this driver to communicate with TJA1145A device.

CanTrcvSpiSequenceName - Reference to a Spi sequence configuration container.

CanTrcvSpiAccessSynchronous - This parameter is used to define whether the access to the Spi sequence is synchronous or asynchronous.

CanTrcvDioChannelAccess (not supported):

CanTrcvHardwareInterfaceName - CAN transceiver hardware interface name. It is typically the name of a pin. From a Dio point of view it is either a port, a single channel or a channel group.

CanTrcvDioSymNameRef - Reference to a DIO Port, DIO Channel or DIO Channel Group.

CanTrcvDemEventParameterRefs:

CANTRCV_E_BUS_ERROR - Reference to the DemEventParameter which shall be issued when bus error has occurred.

CanTrcvPartialNetwork:

CanTrcvBaudRate - Indicates the data transfer rate in kbps.

CanTrcvPnFrameCanId - CAN ID of the Wake-up Frame (WUF).

CanTrcvPnFrameCanIdMask - ID Mask for the selective activation of the transceiver. It is used to enable-Frame Wake-up (WUF) on a group of IDs.

CanTrcvPnFrameDlc - Data Length of the Wake-up Frame (WUF).

CanTrcvFdEnable - Enable/disable flexible data rate.

CanTrcvPnCanIdIsExtended - Indicates whether extended or standard ID is used.

CanTrcvPnEnabled - Indicates whether the selective wake-up function is enabled or disabled in HW.

CanTrcvBusErrFlag - Indicates if the Bus Error (BUSERR) flag is managed by the BSW. This flag is set if a bus failure is detected by the transceiver.

CanTrcvPowerOnFlag - Description: Indicates if the Power On Reset (POR) flag is available and is managed by the transceiver.

WufPattern - Determines whether DLC and data payload will be included in wake-up frame pattern.

CanTrcvPnFrameDataMaskSpec:

CanTrcvPnFrameDataMask - Defines the byte 0 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMask1 - Defines the byte 1 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMask2 - Defines the byte 2 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

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CanTrcvPnFrameDataMask3 - Defines the byte 3 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMask4 - Defines the byte 4 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMask5 - Defines the byte 5 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMask6 - Defines the byte 6 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMask7 - Defines the byte 7 of the data payload mask to be used on the received payload in order to determine if the transceiver must be woken up by the received Wake-up frame (WUF).

CanTrcvPnFrameDataMaskIndex - Holds the position n in frame of the mask-part.

WuConfig:

CanBusWuConfig - Enables/disables wake up by CAN bus.

WakePinConfig - Enable/disable WAKE pin wake up. TJA1145A can be woken up by rising or falling edge of WAKE pin.

EventCaptureConfig:

OvertmpDetection - Enable/disable overtemperature detection.

SpiFailureDetection - Enable/disable SPI failure detection.

CanBusSilenceDetection - Enable/disable CAN bus silence detection.

CanBusFailureDetection - Enable/disable CAN bus failure detection.

vccUvDetection - VCC undervoltage detection in active mode configuration.

LockControl:

LockRange06h09h - Locks register address range from 06h to 09h.

LockRange10h1Fh - Locks register address range from 10h to 1Fh.

LockRange20h2Fh - Locks register address range from 20h to 2Fh.

LockRange30h3Fh - Locks register address range from 30h to 3Fh.

LockRange40h4Fh - Locks register address range from 40h to 4Fh.

LockRange50h5Fh - Locks register address range from 50h to 5Fh.

LockRange68h6Fh - Locks register address range from 68h to 6Fh.

CanTrcv Configuration settings:

HwVersion - Device hardware version.

LeaveMode - Leave configuration mode.

FilterElement - Configuration parameters of the filter elements.



Chapter 5

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5.1 Software Specification

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Chapter 6

Module Documentation

6.1 TJA1145A__DRIVER

6.1.1 Detailed Description

Data Structures

- struct [CanTrcv_43_tja1145a_DriverConfigType](#)
CanTrcv_43_tja1145a_DriverConfigType. More...
- struct [CanTrcv_43_tja1145a_WakeupSourceRefType](#)
Wake up source reference configuration. More...
- struct [CanTrcv_43_tja1145a_EcuMReportType](#)
Stores wake up sources deined by ECU manager. More...
- struct [CanTrcv_tja1145a_SpiConfigType](#)
SPI configuration for access. More...
- struct [CanTrcv_tja1145a_WuConfigType](#)
This data structure contains a configuration of wake functionality. Possible wake up sources are: CAN bus, WAKE pin (active on rising or falling edge). More...
- struct [CanTrcv_tja1145a_EventCaptureConfigType](#)
System/CAN controller configuration structure. Enables/disables various system/CAN controller events. More...
- struct [CanTrcv_tja1145a_CanConfigType](#)
CAN controller configuration structure. More...
- struct [CanTrcv_tja1145a_LockRegistersType](#)
Lock control register configuration. Prevents registers from unintended write access. More...
- struct [CanTrcv_tja1145a_SysDataType](#)
System data of TJA1145A. More...
- struct [CanTrcv_tja1145a_DeviceConfigType](#)
This data structure contains a configuration of the TJA1145A device. This structure is used as a parameter for TJA1145A_Init function. More...
- struct [CanTrcv_tja1145a_ConfigurationDataType](#)
Contains a reference to configuration of TJA1145A and a reference to configuration of SPI peripheral through which the driver communicates with the device. More...
- struct [CanTrcv_tja1145a_DriverDataType](#)
Stores information about device's/driver's status and configuration. More...

Macros

- `#define TJA1145A_M_MC_MASK`
- `#define TJA1145A_M_MC_SHIFT`
- `#define TJA1145A_M_MC_SLEEP_MODE`
- `#define TJA1145A_M_MC_STANDBY_MODE`
- `#define TJA1145A_M_MC_NORMAL_MODE`
- `#define TJA1145A_M_NMS_MASK`
- `#define TJA1145A_M_OTWS_MASK`
- `#define TJA1145A_M_FSMS_MASK`
- `#define TJA1145A_M_NMS_SHIFT`
- `#define TJA1145A_M_OTWS_SHIFT`
- `#define TJA1145A_M_FSMS_SHIFT`
- `#define TJA1145A_M_NMS_WAS_OR_IS_IN_NORMAL_MODE`
- `#define TJA1145A_M_NMS_HAS_NOT_BEEN_IN_NORMAL_MODE`
- `#define TJA1145A_M_OTWS_TEMPERATURE_BELOW_WARNING_THRESHOLD`
- `#define TJA1145A_M_OTWS_TEMPERATURE_ABOVE_WARNING_THRESHOLD`
- `#define TJA1145A_M_FSMS_SLEEP_MODE_REASON_SPI_COMMAND`
- `#define TJA1145A_M_FSMS_SLEEP_MODE_REASON_UNDERVOLTAGE`
- `#define TJA1145A_M_SPIFE_MASK`
- `#define TJA1145A_M_OTWE_MASK`
- `#define TJA1145A_M_SPIFE_SHIFT`
- `#define TJA1145A_M_OTWE_SHIFT`
- `#define TJA1145A_M_SPIFE_DISABLED`
- `#define TJA1145A_M_SPIFE_ENABLED`
- `#define TJA1145A_M_OTWE_DISABLED`
- `#define TJA1145A_M_OTWE_ENABLED`
- `#define TJA1145A_M_GPM_MASK`
- `#define TJA1145A_M_GPM_MASK`
- `#define TJA1145A_M_GPM_MASK`
- `#define TJA1145A_M_GPM_MASK`
- `#define TJA1145A_M_GPM_SHIFT`
- `#define TJA1145A_M_GPM_SHIFT`
- `#define TJA1145A_M_GPM_SHIFT`
- `#define TJA1145A_M_GPM_SHIFT`
- `#define TJA1145A_M_LK0C_MASK`
- `#define TJA1145A_M_LK1C_MASK`
- `#define TJA1145A_M_LK2C_MASK`
- `#define TJA1145A_M_LK3C_MASK`
- `#define TJA1145A_M_LK4C_MASK`
- `#define TJA1145A_M_LK5C_MASK`
- `#define TJA1145A_M_LK6C_MASK`
- `#define TJA1145A_M_LK0C_SHIFT`
- `#define TJA1145A_M_LK1C_SHIFT`
- `#define TJA1145A_M_LK2C_SHIFT`
- `#define TJA1145A_M_LK3C_SHIFT`
- `#define TJA1145A_M_LK4C_SHIFT`
- `#define TJA1145A_M_LK5C_SHIFT`
- `#define TJA1145A_M_LK6C_SHIFT`
- `#define TJA1145A_M_LK0C_WRITING_ENABLED`

- #define TJA1145A_M_LK0C_WRITING_DISABLED
- #define TJA1145A_M_LK1C_WRITING_ENABLED
- #define TJA1145A_M_LK1C_WRITING_DISABLED
- #define TJA1145A_M_LK2C_WRITING_ENABLED
- #define TJA1145A_M_LK2C_WRITING_DISABLED
- #define TJA1145A_M_LK3C_WRITING_ENABLED
- #define TJA1145A_M_LK3C_WRITING_DISABLED
- #define TJA1145A_M_LK4C_WRITING_ENABLED
- #define TJA1145A_M_LK4C_WRITING_DISABLED
- #define TJA1145A_M_LK5C_WRITING_ENABLED
- #define TJA1145A_M_LK5C_WRITING_DISABLED
- #define TJA1145A_M_LK6C_WRITING_ENABLED
- #define TJA1145A_M_LK6C_WRITING_DISABLED
- #define TJA1145A_FS_I_CMC_MASK
- #define TJA1145A_FS_I_CPNC_MASK
- #define TJA1145A_FS_I_PNCOK_MASK
- #define TJA1145A_FS_I_CFDC_MASK
- #define TJA1145A_FS_I_CMC_SHIFT
- #define TJA1145A_FS_I_CPNC_SHIFT
- #define TJA1145A_FS_I_PNCOK_SHIFT
- #define TJA1145A_FS_I_CFDC_SHIFT
- #define TJA1145A_FS_I_CMC_OFFLINE_MODE
- #define TJA1145A_FS_I_CMC_ACTIVE_MODE__01
- #define TJA1145A_FS_I_CMC_ACTIVE_MODE__10
- #define TJA1145A_FS_I_CMC_LISTEN_ONLY_MODE
- #define TJA1145A_FS_I_CPNC_SELECTIVE_WAKE_DISABLED
- #define TJA1145A_FS_I_CPNC_SELECTIVE_WAKE_ENABLED
- #define TJA1145A_FS_I_PNCOK_PN_CONFIGURED
- #define TJA1145A_FS_I_CFDC_FD_DISABLED
- #define TJA1145A_FS_I_CFDC_FD_ENABLED
- #define TJA1145A_FS_I_CFS_MASK
- #define TJA1145A_FS_I_VCS_MASK
- #define TJA1145A_FS_I_CBSS_MASK
- #define TJA1145A_FS_I_COSCS_MASK
- #define TJA1145A_FS_I_CPNS_MASK
- #define TJA1145A_FS_I_CPNERR_MASK
- #define TJA1145A_FS_I_CTS_MASK
- #define TJA1145A_FS_I_CFS_SHIFT
- #define TJA1145A_FS_I_VCS_SHIFT
- #define TJA1145A_FS_I_CBSS_SHIFT
- #define TJA1145A_FS_I_COSCS_SHIFT
- #define TJA1145A_FS_I_CPNS_SHIFT
- #define TJA1145A_FS_I_CPNERR_SHIFT
- #define TJA1145A_FS_I_CTS_SHIFT
- #define TJA1145A_FS_I_CFS_NO_TIME_OUT_EVENT_DETECTED
- #define TJA1145A_FS_I_CFS_TIME_OUT_EVENT_DETECTED
- #define TJA1145A_FS_I_VCS_ABOVE_UNDERVOLTAGE_THRESHOLD
- #define TJA1145A_FS_I_VCS_BELOW_UNDERVOLTAGE_THRESHOLD
- #define TJA1145A_FS_I_CBSS_CAN_BUS_ACTIVE
- #define TJA1145A_FS_I_CBSS_CAN_BUS_INACTIVE

- #define TJA1145A_FS_I_COSCS_NOT_AT_TARGET_FREQUENCY
- #define TJA1145A_FS_I_COSCS_AT_TARGET_FREQUENCY
- #define TJA1145A_FS_I_CPNS_CONFIGURATION_ERROR_DETECTED
- #define TJA1145A_FS_I_CPNS_CONFIGURATION_OK
- #define TJA1145A_FS_I_CPNERR_NO_ERROR_DETECTED
- #define TJA1145A_FS_I_CPNERR_ERROR_DETECTED
- #define TJA1145A_FS_I_CTS_NOT_IN_ACTIVE_MODE
- #define TJA1145A_FS_I_CTS_IN_ACTIVE_MODE
- #define TJA1145A_FS_I_CWE_MASK
- #define TJA1145A_FS_I_CFE_MASK
- #define TJA1145A_FS_I_CBSE_MASK
- #define TJA1145A_FS_I_CWE_SHIFT
- #define TJA1145A_FS_I_CFE_SHIFT
- #define TJA1145A_FS_I_CBSE_SHIFT
- #define TJA1145A_FS_I_CWE_DISABLED
- #define TJA1145A_FS_I_CWE_ENABLED
- #define TJA1145A_FS_I_CFE_DISABLED
- #define TJA1145A_FS_I_CFE_ENABLED
- #define TJA1145A_FS_I_CBSE_DISABLED
- #define TJA1145A_FS_I_CBSE_ENABLED
- #define TJA1145A_FS_I_CDR_MASK
- #define TJA1145A_FS_I_CDR_SHIFT
- #define TJA1145A_FS_I_CDR_50_KBIT_S
- #define TJA1145A_FS_I_CDR_100_KBIT_S
- #define TJA1145A_FS_I_CDR_125_KBIT_S
- #define TJA1145A_FS_I_CDR_250_KBIT_S
- #define TJA1145A_FS_I_CDR_500_KBIT_S
- #define TJA1145A_FS_I_CDR_1000_KBIT_S
- #define TJA1145A_FS_I_ID0_MASK
- #define TJA1145A_FS_I_ID0_SHIFT
- #define TJA1145A_FS_I_M_MASK
- #define TJA1145A_FS_I_M_SHIFT
- #define TJA1145A_FS_I_DLC_MASK
- #define TJA1145A_FS_I_PNDM_MASK
- #define TJA1145A_FS_I_IDE_MASK
- #define TJA1145A_FS_I_DLC_SHIFT
- #define TJA1145A_FS_I_PNDM_SHIFT
- #define TJA1145A_FS_I_IDE_SHIFT
- #define TJA1145A_FS_I_DLC_0
- #define TJA1145A_FS_I_DLC_1
- #define TJA1145A_FS_I_DLC_2
- #define TJA1145A_FS_I_DLC_3
- #define TJA1145A_FS_I_DLC_4
- #define TJA1145A_FS_I_DLC_5
- #define TJA1145A_FS_I_DLC_6
- #define TJA1145A_FS_I_DLC_7
- #define TJA1145A_FS_I_DLC_8
- #define TJA1145A_FS_I_DLC_1001__8_BYTES_EXPECTED__
- #define TJA1145A_FS_I_DLC_1010__8_BYTES_EXPECTED__
- #define TJA1145A_FS_I_DLC_1011__8_BYTES_EXPECTED__

- #define TJA1145A_FS_I_DLC_1100__8_BYTES_EXPECTED__
- #define TJA1145A_FS_I_DLC_1101__8_BYTES_EXPECTED__
- #define TJA1145A_FS_I_DLC_1110__8_BYTES_EXPECTED__
- #define TJA1145A_FS_I_DLC_1111__8_BYTES_EXPECTED__
- #define TJA1145A_FS_I_PNDM_DLC__DATA__DON_T_CARE__
- #define TJA1145A_FS_I_PNDM_DLC__DATA_EVALUATED
- #define TJA1145A_FS_I_IDE_STANDARD__11_BIT__
- #define TJA1145A_FS_I_IDE_EXTENDED__29_BIT__
- #define TJA1145A_FS_I_DM0_MASK
- #define TJA1145A_FS_I_DM0_SHIFT
- #define TJA1145A_FS_I_DM1_MASK
- #define TJA1145A_FS_I_DM1_SHIFT
- #define TJA1145A_FS_I_DM2_MASK
- #define TJA1145A_FS_I_DM2_SHIFT
- #define TJA1145A_FS_I_DM3_MASK
- #define TJA1145A_FS_I_DM3_SHIFT
- #define TJA1145A_FS_I_DM4_MASK
- #define TJA1145A_FS_I_DM4_SHIFT
- #define TJA1145A_FS_I_DM5_MASK
- #define TJA1145A_FS_I_DM5_SHIFT
- #define TJA1145A_FS_I_DM6_MASK
- #define TJA1145A_FS_I_DM6_SHIFT
- #define TJA1145A_FS_I_DM7_MASK
- #define TJA1145A_FS_I_DM7_SHIFT
- #define TJA1145A_FS_WPVS_MASK
- #define TJA1145A_FS_WPVS_SHIFT
- #define TJA1145A_FS_WPVS_BELOW_SWITCHING_THRESHOLD
- #define TJA1145A_FS_WPVS_ABOVE_SWITCHING_THRESHOLD
- #define TJA1145A_FS_WPFE_MASK
- #define TJA1145A_FS_WPRE_MASK
- #define TJA1145A_FS_WPFE_SHIFT
- #define TJA1145A_FS_WPRE_SHIFT
- #define TJA1145A_FS_WPFE_FALLING_EDGE_DISABLED
- #define TJA1145A_FS_WPFE_FALLING_EDGE_ENABLED
- #define TJA1145A_FS_WPRE_RISING_EDGE_DISABLED
- #define TJA1145A_FS_WPRE_RISING_EDGE_ENABLED
- #define TJA1145A_FS_SYSE_MASK
- #define TJA1145A_FS_TRXE_MASK
- #define TJA1145A_FS_WPE_MASK
- #define TJA1145A_FS_SYSE_SHIFT
- #define TJA1145A_FS_TRXE_SHIFT
- #define TJA1145A_FS_WPE_SHIFT
- #define TJA1145A_FS_SYSE_NO_PENDING_EVENTS
- #define TJA1145A_FS_SYSE_PENDING_EVENT_S__
- #define TJA1145A_FS_TRXE_NO_PENDING_EVENTS
- #define TJA1145A_FS_TRXE_PENDING_EVENT_S__
- #define TJA1145A_FS_WPE_NO_PENDING_EVENTS
- #define TJA1145A_FS_WPE_PENDING_EVENT_S__
- #define TJA1145A_FS_SPIF_MASK
- #define TJA1145A_FS_OTW_MASK

- `#define TJA1145A_FS_PO_MASK`
- `#define TJA1145A_FS_SPIF_SHIFT`
- `#define TJA1145A_FS_OTW_SHIFT`
- `#define TJA1145A_FS_PO_SHIFT`
- `#define TJA1145A_FS_SPIF_SPI_FAILURE_EVENT`
- `#define TJA1145A_FS_OTW_OVERTEMP_WARNING_EVENT`
- `#define TJA1145A_FS_PO_POWER_ON_EVENT`
- `#define TJA1145A_FS_CW_MASK`
- `#define TJA1145A_FS_CF_MASK`
- `#define TJA1145A_FS_CBS_MASK`
- `#define TJA1145A_FS_PNFDE_MASK`
- `#define TJA1145A_FS_CW_SHIFT`
- `#define TJA1145A_FS_CF_SHIFT`
- `#define TJA1145A_FS_CBS_SHIFT`
- `#define TJA1145A_FS_PNFDE_SHIFT`
- `#define TJA1145A_FS_CW_CAN_WAKE_UP_EVENT`
- `#define TJA1145A_FS_CF_CAN_FAILURE_EVENT`
- `#define TJA1145A_FS_CBS_BUS_SILENCE_EVENT`
- `#define TJA1145A_FS_PNFDE_FRAME_ERROR_EVENT`
- `#define TJA1145A_FS_WPF_MASK`
- `#define TJA1145A_FS_WPR_MASK`
- `#define TJA1145A_FS_WPF_SHIFT`
- `#define TJA1145A_FS_WPR_SHIFT`
- `#define TJA1145A_FS_WPF_FALLING_EDGE_EVENT`
- `#define TJA1145A_FS_WPR_RISING_EDGE_EVENT`
- `#define TJA1145A_FS_IDS_MASK`
- `#define TJA1145A_FS_IDS_SHIFT`
- `#define TJA1145A_FS_IDS_TJA1145AATK`
- `#define TJA1145A_FS_IDS_TJA1145AATK_FD`

Enum Reference

- enum `CanTrcv_43_tja1145a_eDriverStatusType`
CanTrcv_43_tja1145a_eDriverStatusType.
- enum `CanTrcv_tja1145a_StatusType`
Enumeration of possible results of command execution.
- enum `CanTrcv_tja1145a_SystemErrorType`
System error.
- enum `CanTrcv_tja1145a_PorEventType`
Power on event flag.
- enum `CanTrcv_tja1145a_WuReasonType`
Wake up source.
- enum `CanTrcv_tja1145a_WuPinModeType`
WAKE pin wake up feature mode.
- enum `CanTrcv_tja1145a_WuCanModeType`
CAN bus wake up feature mode.
- enum `CanTrcv_tja1145a_CanDataRateType`
CAN data rate.

- enum [CanTrcv_tja1145a_IdFormatType](#)
Wake up frame identifier.
- enum [CanTrcv_tja1145a_WufEvalType](#)
There are 2 options of how a WUF can be evaluated:
- enum [CanTrcv_tja1145a_CanControllerModeType](#)
CAN controller modes.
- enum [CanTrcv_tja1145a_FdModeType](#)
Flexible data rate (available for TJA1145AT/FD and TJA1145ATK/FD only).
- enum [CanTrcv_tja1145a_VccUvConfigType](#)
VCC undervoltage detection in CAN active mode configuration. If enabled CAN controller will automatically switch to offline mode if VCC undervoltage is detected.
- enum [CanTrcv_tja1145a_PNActivationType](#)
Partial networking configuration.
- enum [CanTrcv_tja1145a_TrcvFlagStateType](#)
Indicates flag's state.
- enum [CanTrcv_tja1145a_MemoryRegisterType](#)
List of general purpose memory registers.

Function Reference

- void [CanTrcv_43_tja1145a_Init](#) (const [CanTrcv_43_tja1145a_DriverConfigType](#) *ConfigPtr)
Initializes CanTrcv module. SID 0x00.
- Std_ReturnType [CanTrcv_43_tja1145a_SetOpMode](#) (uint8 Transceiver, CanTrcv_TrcvModeType OpMode)
Sets the mode of the Transceiver to the value OpMode. SID 0x01.
- Std_ReturnType [CanTrcv_43_tja1145a_GetOpMode](#) (uint8 Transceiver, CanTrcv_TrcvModeType *OpMode)
Gets current operation mode. SID 0x02.
- Std_ReturnType [CanTrcv_43_tja1145a_GetBusWuReason](#) (uint8 Transceiver, CanTrcv_TrcvWakeupReasonType *Reason)
Gets the wakeup reason of the Transceiver and returns it in parameter Reason. SID 0x03.
- Std_ReturnType [CanTrcv_43_tja1145a_GetTrcvSystemData](#) (uint8 Transceiver, uint32 *TrcvSysData)
Reads transceiver system data. SID 0x09.
- Std_ReturnType [CanTrcv_43_tja1145a_ClearTrcvWufFlag](#) (uint8 Transceiver)
Clears the WUF flag in the transceiver hardware. SID 0x0A.
- Std_ReturnType [CanTrcv_43_tja1145a_SetWakeupMode](#) (uint8 Transceiver, CanTrcv_TrcvWakeupModeType TrcvWakeupMode)
Enables, disables or clears wake up events of the Transceiver according to TrcvWakeupMode. SID 0x05.
- Std_ReturnType [CanTrcv_43_tja1145a_CheckWakeup](#) (uint8 Transceiver)
Service is called by underlying CANIF in case a wake up interrupt is detected. SID 0x07.
- Std_ReturnType [CanTrcv_43_tja1145a_CheckWakeFlag](#) (uint8 Transceiver)
Requests to check the status of the wakeup flag from the transceiver hardware. SID 0x0E.
- Std_ReturnType [CanTrcv_43_tja1145a_ReadTrcvTimeoutFlag](#) (uint8 Transceiver, [CanTrcv_tja1145a_TrcvFlagStateType](#) *FlagState)
Reads the status of the timeout flag from the transceiver hardware. SID 0x0B.
- Std_ReturnType [CanTrcv_43_tja1145a_ClearTrcvTimeoutFlag](#) (uint8 Transceiver)
Clears the status of the timeout flag in the transceiver hardware. SID 0x0C.

- Std_ReturnType [CanTrcv_43_tja1145a_ReadTrcvSilenceFlag](#) (uint8 Transceiver, [CanTrcv_tja1145a_TrvcFlagStateType](#) *FlagState)
Reads the status of the silence flag from the transceiver hardware. SID 0x0D.
- Std_ReturnType [CanTrcv_43_tja1145a_SetPNActivationState](#) ([CanTrcv_tja1145a_PNActivationType](#) ActivationState)
Sets PN functionality state. SID 0x0F.
- Std_ReturnType [CanTrcv_43_tja1145a_Exts_ClearPorFlag](#) (uint8 Transceiver)
Clears POR flag.
- Std_ReturnType [CanTrcv_43_tja1145a_Exts_ClearSystemError](#) (uint8 Transceiver, [CanTrcv_tja1145a_SystemErrorType](#) SystemError)
Clears system error.
- Std_ReturnType [CanTrcv_43_tja1145a_Exts_WriteMemoryRegister](#) (uint8 Transceiver, [CanTrcv_tja1145a_MemoryRegister](#) MemRegister, uint8 Data)
Writes general purpose memory register.
- Std_ReturnType [CanTrcv_43_tja1145a_Exts_ReadMemoryRegister](#) (uint8 Transceiver, [CanTrcv_tja1145a_MemoryRegister](#) MemRegister, uint8 *Data)
Reads general purpose memory register.
- Std_ReturnType [CanTrcv_43_tja1145a_Exts_SetLockControlRegister](#) (uint8 Transceiver, const [CanTrcv_tja1145a_LockControlType](#) *LockControl)
Locks/unlocks register groups.
- void [CanTrcv_43_tja1145a_MainFunction](#) (void)
Service to scan all busses for wake up events and perform these event. SID 0x06.
- void [CanTrcv_43_tja1145a_MainFunctionDiagnostics](#) (void)
Reads the transceiver diagnostic status periodically and sets product/development accordingly. SID 0x08.
- void [CanTrcv_43_tja1145a_DeInit](#) (void)
De-initializes the CanTrcv module. SID 0x10.
- void [TJA1145A_DRV_Init](#) (uint8 Transceiver, const [CanTrcv_tja1145a_ConfigurationDataType](#) *ConfigData)
Initializes the TJA1145A driver. After driver initialization, function TJA1145A_DRV_ConfigureDevice must be called to configure the transceiver and finalize overall configuration. Having driver and device configuration in separate functions enables TJA1145A reconfiguration on cold start (MCU was off) only if POR or error event has been detected.
- [CanTrcv_tja1145a_StatusType](#) [TJA1145A_DRV_ConfigureDevice](#) (uint8 Transceiver, const [CanTrcv_tja1145a_DeviceConfigType](#) *DeviceConfig)
Configures the transceiver. Writes data from configuration structure to its registers.
- [CanTrcv_tja1145a_StatusType](#) [TJA1145A_DRV_ReadRegister](#) (uint8 Transceiver, uint8 RegAddress, uint8 *RxData)
Performs a read from a single TJA1145A register. Reads a single TJA1145A register. Read out data is stored in a location pointed by pRxData.
- [CanTrcv_tja1145a_StatusType](#) [TJA1145A_DRV_ReadMultipleRegisters](#) (uint8 Transceiver, uint8 RegAddress, uint16 NumRegisters, uint8 *RxData)
Performs a read from multiple TJA1145A registers. Reads up to three consecutive TJA1145A registers. Address of the first register must be provided via the 'Address' parameter. TJA1145A will then automatically increment addresses of following registers. Read out data is stored in a location pointed by pRxData. Note: it is important to allocate enough space to accommodate all read out data.
- [CanTrcv_tja1145a_StatusType](#) [TJA1145A_DRV_WriteRegister](#) (uint8 Transceiver, uint8 RegAddress, uint8 TxData)
Writes a single TJA1145A register.
- [CanTrcv_tja1145a_StatusType](#) [TJA1145A_DRV_WriteMultipleRegisters](#) (uint8 Transceiver, uint8 RegAddress, uint16 NumRegisters, const uint8 *TxData)

Writes multiple TJA1145A consecutive registers. Writes up to the three consecutive TJA1145A registers. The address of the first register must be provided via the 'Address' parameter. Addresses of following registers are automatically incremented by TJA1145A.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_UpdateRegister](#) (uint8 Transceiver, uint8 RegAddress, uint8 Value, uint8 Mask, uint8 Shift)

Updates a bit/bit group in a register and leaves the rest of the register unchanged.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetMode](#) (uint8 Transceiver, CanTrcv_TrcvModeType OpMode)

Sets the mode of the transceiver to the Value OpMode.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetMode](#) (uint8 Transceiver, CanTrcv_TrcvModeType *OpMode)

Reads the mode of the transceiver from 'Mode control register' and returns it in OpMode parameter.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetWuReason](#) (uint8 Transceiver, CanTrcv_tja1145a_WuReasonType *WuReason)

Retrieves the wake up reason. Possible wake up sources are: CAN bus, wake pin (either rising or falling edge).

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_EventCaptureConfiguration](#) (uint8 Transceiver, const CanTrcv_tja1145a_EventCaptureConfigType *EventConfig)

Writes system and CAN controller event capture configuration.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_WakeupModeConfig](#) (uint8 Transceiver, boolean WakeEnable)

Enables/disables wake up mode according to WakeEnable parameter.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_WakeupConfiguration](#) (uint8 Transceiver, const CanTrcv_tja1145a_Wu *WakeupConfig)

Enables/disables WAKE pin and CAN bus wake up feature.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetPnMode](#) (uint8 Transceiver, CanTrcv_tja1145a_PNActivationType PnMode)

Enables/disables partial networking functionality. If PN is activated, the device can be woken up by a remote frame. If PN is disabled, the device can be woken up only by a standard wake up pattern. It is expected that partial networking has been configured prior calling this function which only enables/disables the PN functionality.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetCanConfig](#) (uint8 Transceiver, const CanTrcv_tja1145a_CanConfig *CanConfig)

Configures CAN controller.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetCanControllerMode](#) (uint8 Transceiver, CanTrcv_tja1145a_CanControllerMode CanMode)

Sets mode of CAN controller which may differ from mode of TJA1145A. These are two entities, each with its own mode.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadTimeoutFlag](#) (uint8 Transceiver, CanTrcv_tja1145a_TrcvFlagState *FlagState)

Reads timeout flag.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadSilenceFlag](#) (uint8 Transceiver, CanTrcv_tja1145a_TrcvFlagState *FlagState)

Reads silence flag.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetTrcvSystemData](#) (uint8 Transceiver, CanTrcv_tja1145a_SysDataT *TrcvSysData)

Returns status of the transceiver (its mode, overtemperature flag) and status of CAN controller (PN status, CAN bus status, ...).

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_ClearTrcvWufFlag](#) (uint8 Transceiver, CanTrcv_tja1145a_WuReasonT WuReason)

Clears wake up flag. Clears one of following wake up flags (interrupts can as well wake up the device): WAKE pin flag, CAN bus wake up flag, SPI failure (this interrupt can wake up the device only when it is in standby mode), partial networking frame error, CAN bus silence.

- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetPorEventStatus](#) (uint8 Transceiver, [CanTrcv_tja1145a_PorEventT](#), *PorEventStatus)
Reads power on event flag.
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_ClearPorFlag](#) (uint8 Transceiver)
Clears power on event flag.
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetSystemErrorStatus](#) (uint8 Transceiver, uint16 *SystemError)
Returns system error status. This driver recognizes following system errors:
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_ClearSystemErrorStatus](#) (uint8 Transceiver, [CanTrcv_tja1145a_SystemErrorT](#), SystemError)
Clears system error. Important: it is not possible to clear all system errors. Following system errors are read only flags: CAN oscillator status, VCC undervoltage, PN configuration error (in this case PN must be reconfigured). Note: TJA1145A_PN_ERROR flag can be set because of two reason:
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetLockControlRegister](#) (uint8 Transceiver, const [CanTrcv_tja1145a_LockRegistersType](#) *LockControl)
Writes lock configuration to prevent unintended register access.
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_WriteMemoryRegister](#) (uint8 Transceiver, [CanTrcv_tja1145a_MemoryRegistersType](#) MemRegister, uint8 Data)
Writes general purpose memory register.
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadMemoryRegister](#) (uint8 Transceiver, [CanTrcv_tja1145a_MemoryRegistersType](#) MemRegister, uint8 *Data)
Reads general purpose memory register.
- [CanTrcv_tja1145a_StatusType TJA1145A_DRV_Deinit](#) (uint8 Transceiver)
Deinitializes the TJA1145A driver. SID is 0x0F. Clears the internal run-time configuration and puts the device to standby mode.

Variables

- uint8 [DeviceNum](#)
Number of devices controlled by the driver. The data type is uint8, because Autosar API indexes connected devices with an index of type uint8.
- const [CanTrcv_tja1145a_ConfigurationDataType](#) * [ConfigData](#)
Pointer to an array of device and SPI configuration structures.
- boolean [WakeupSourceEn](#)
Wake up source enabled/disabled.
- EcuM_WakeupSourceType [EcuMWakeupSource](#)
Reference to the wake up source defined in the ECU state manager.
- [CanTrcv_43_tja1145a_WakeupSourceRefType](#) [WakeupSourceConfig](#)
Configuration of standard wake up source used in reports to ECU manager.
- [CanTrcv_43_tja1145a_WakeupSourceRefType](#) [PorWakeupSourceConfig](#)
Configuration of POR wake up source used in reports to ECU manager.
- [CanTrcv_43_tja1145a_WakeupSourceRefType](#) [SysErrWakeupSourceConfig](#)
Configuration of SYSERR wake up source used in reports to ECU manager.
- Mcal_DemErrorType [DemErrorReportConfig](#)
DEM reporting configuration.
- volatile [CanTrcv_43_tja1145a_eDriverStatusType](#) [CanTrcv_43_tja1145a_eDriverStatus](#)
Holds current status/run-time of CANTRCV driver: CANTRCV_43_TJA1145A_UNINIT or CANTRCV_43_TJA1145A_READY.

6.1.1.1 MISRA-C:2012 violations

6.1.2 Data Structure Documentation

6.1.2.1 struct CanTrcv_43_tja1145a_DriverConfigType

[CanTrcv_43_tja1145a_DriverConfigType](#).

Externally defined data structure containing the overall initialization data for the CAN transceiver driver and settings of all transceivers.

Definition at line 207 of file [CanTrcv_43_tja1145a.h](#).

Data Fields

Type	Name	Description
uint8	DeviceNum	Number of devices controlled by the driver. The data type is uint8, because Autosar API indexes connected devices with an index of type uint8.
const CanTrcv_tja1145a_ConfigurationDataType *	ConfigData	Pointer to an array of device and SPI configuration structures.

6.1.2.2 struct CanTrcv_43_tja1145a_WakeupSourceRefType

Wake up source reference configuration.

ECU manager defines wake up sources (POR, standard wake up and system error). This structure stores configuration of such wake up source.

Definition at line 221 of file [CanTrcv_43_tja1145a.h](#).

Data Fields

Type	Name	Description
boolean	WakeupSourceEn	Wake up source enabled/disabled.
EcuM_WakeupSourceType	EcuMWakeupSource	Reference to the wake up source defined in the ECU state manager.

6.1.2.3 struct CanTrcv_43_tja1145a_EcuMReportType

Stores wake up sources defined by ECU manager.

ECU manager defines wake up sources which are used by the driver for reporting wake up events to the ECU state manager.

Definition at line [234](#) of file [CanTrcv_43_tja1145a.h](#).

Data Fields

Type	Name	Description
CanTrcv_43_tja1145a_WakeupSourceRefType	WakeupSourceConfig	Configuration of standard wake up source used in reports to ECU manager.
CanTrcv_43_tja1145a_WakeupSourceRefType	PorWakeupSourceConfig	Configuration of POR wake up source used in reports to ECU manager.
CanTrcv_43_tja1145a_WakeupSourceRefType	SysErrWakeupSourceConfig	Configuration of SYSERR wake up source used in reports to ECU manager.
Mcal_DemErrorType	DemErrorReportConfig	DEM reporting configuration.

6.1.2.4 struct CanTrcv_tja1145a_SpiConfigType

SPI configuration for access.

Definition at line 407 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
uint32	SpiChannel	SPI instance (non-AUTOSAR) / SPI channel (AUTOSAR).
const void *	SpiExtension	SPI External Device (non-AUTOSAR) / SPI sequence (AUTOSAR).

6.1.2.5 struct CanTrcv_tja1145a_WuConfigType

This data structure contains a configuration of wake functionality. Possible wake up sources are: CAN bus, WAKE pin (active on rising or falling edge).

Definition at line 420 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
CanTrcv_tja1145a_WuCanModeType	CanBusWuMode	CAN bus wake up functionality mode (enabled/disabled)
CanTrcv_tja1145a_WuPinModeType	WuPinMode	WAKE pin wake up mode (disabled/active on rising edge/active on falling edge).

6.1.2.6 struct CanTrcv_tja1145a_EventCaptureConfigType

System/CAN controller configuration structure. Enables/disables various system/CAN controller events.

Definition at line 433 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
boolean	OvertempDetectionEnable	Overtemperature warning configuration.
boolean	SpiFailureDetectionEnable	SPI failure configuration.
boolean	CanBusSilenceDetectionEnable	CAN bus silence configuration.
boolean	CanFailureDetectionEnable	CAN failure warning configuration.

6.1.2.7 struct CanTrcv_tja1145a_CanConfigType

CAN controller configuration structure.

Definition at line 449 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
uint32	WufId	ID of wake up frame (WUF). Value stored in this item defines WUF ID which will be considered a valid wake up ID. 'WufId' in combination with 'WufIdMask' defines a range a valid IDs.
uint32	WufIdMask	ID Mask of wake up frame (WUF). 1s in this item mean 'don't care' state. For example a Mask b00000111 allows a range of 8 possible valid WUF IDs. See datasheet 'Wake-up frame' section for more details.
uint8	Dlc	Data length code. The 'Dlc' item indicates how many data bytes (0 - 8 bytes) are expected in the data field of a wake up frame (WUF). If 'Dlc' = 0, a node will wake up if the WUF contains a valid identifier and length of received data is 0.
CanTrcv_tja1145a_CanDataRateType	DataRate	Data rate selection.
CanTrcv_tja1145a_WufEvalType	WufEval	Determines whether only ID is considered in WUF evaluation or also DLC and data bytes along with ID are part of WUF evaluation.
CanTrcv_tja1145a_IdFormatType	IdFormat	ID format: 11 or 29 bit ID format.
uint8	DataMasks[8U]	Data masks.
CanTrcv_tja1145a_PNActivationType	PnMode	Partial networking mode (enable/disable).
CanTrcv_tja1145a_FdModeType	FdConfig	Flexible data rate configuration.

6.1.2.8 struct CanTrcv_tja1145a_LockRegistersType

Lock control register configuration. Prevents registers from unintended write access.

Definition at line 483 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
boolean	LockRange06h09h	Locks register within address range 06h - 09h.
boolean	LockRange10h1Fh	Locks register within address range 10h - 1Fh.
boolean	LockRange20h2Fh	Locks register within address range 20h - 2Fh.
boolean	LockRange30h3Fh	Locks register within address range 30h - 3Fh.
boolean	LockRange40h4Fh	Locks register within address range 40h - 4Fh.
boolean	LockRange50h5Fh	Locks register within address range 50h - 5Fh.
boolean	LockRange68h6Fh	Locks register within address range 68h - 6Fh.

6.1.2.9 struct CanTrcv_tja1145a_SysDataType

System data of TJA1145A.

Definition at line 505 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
uint8	MainStatusReg	Main status register content.
uint8	TrcvStatusReg	Transceiver status register content.
uint8	GlobalEventStatusReg	Global event status register content.
uint8	SysEventStatusReg	System event status register content.
uint8	TrcvEventStatusReg	Transceiver event status register content.
uint8	WakePinEventStatusReg	WAKE pin event status register content.

6.1.2.10 struct CanTrcv_tja1145a_DeviceConfigType

This data structure contains a configuration of the TJA1145A device. This structure is used as a parameter for TJA1145A_Init function.

Definition at line 527 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

- [CanTrcv_tja1145a_CanConfigType CanConfig](#)
CAN controller configuration.
- [CanTrcv_tja1145a_WuConfigType WuConfig](#)
Wake up functionality configuration.
- [CanTrcv_tja1145a_EventCaptureConfigType EventConfig](#)

System/CAN controller event configuration.

- [CanTrcv_tja1145a_VccUvConfigType](#) [VccUvConfig](#)
VCC undervoltage detection in active mode configuration.
- [CanTrcv_tja1145a_LockRegistersType](#) [LockControl](#)
Lock control register configuration.
- [CanTrcv_TrcvModeType](#) [InitState](#)
Initial state.

6.1.2.10.1 Field Documentation

6.1.2.10.1.1 CanConfig [CanTrcv_tja1145a_CanConfigType](#) [CanConfig](#)

CAN controller configuration.

Definition at line 530 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.2.10.1.2 WuConfig [CanTrcv_tja1145a_WuConfigType](#) [WuConfig](#)

Wake up functionality configuration.

Definition at line 532 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.2.10.1.3 EventConfig [CanTrcv_tja1145a_EventCaptureConfigType](#) [EventConfig](#)

System/CAN controller event configuration.

Definition at line 534 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.2.10.1.4 VccUvConfig [CanTrcv_tja1145a_VccUvConfigType](#) [VccUvConfig](#)

VCC undervoltage detection in active mode configuration.

Definition at line 536 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.2.10.1.5 LockControl [CanTrcv_tja1145a_LockRegistersType](#) [LockControl](#)

Lock control register configuration.

Definition at line 538 of file [CanTrcv_tja1145a_Ip.h](#).

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6.1.2.10.1.6 InitState CanTrcv_TrvcModeType InitState

Initial state.

Definition at line 540 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.2.11 struct CanTrcv_tja1145a_ConfigurationDataType

Contains a reference to configuration of TJA1145A and a reference to configuration of SPI peripheral through which the driver communicates with the device.

Definition at line 548 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
const CanTrcv_tja1145a_SpiConfigType *	SpiConfig	Pointer to SPI configuration.
const CanTrcv_tja1145a_DeviceConfigType *	DeviceConfig	Pointer to an array of device configuration structure.

6.1.2.12 struct CanTrcv_tja1145a_DriverDataType

Stores information about device's/driver's status and configuration.

Definition at line 560 of file [CanTrcv_tja1145a_Ip.h](#).

Data Fields

Type	Name	Description
const CanTrcv_tja1145a_SpiConfigType *	SpiConfig	Pointer to SPI configuration.
const CanTrcv_tja1145a_DeviceConfigType *	DeviceConfig	Pointer to an array of device configuration structure.
CanTrcv_tja1145a_WuReasonType	WuReason	Report of a wake up reason.
uint16	SystemError	Report of system error(s).

6.1.3 Macro Definition Documentation

6.1.3.1 TJA1145A_M_MC_MASK

```
#define TJA1145A_M_MC_MASK
```

mode control

Definition at line 98 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.2 TJA1145A_M_MC_SHIFT

```
#define TJA1145A_M_MC_SHIFT
```

mode control

Definition at line 103 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.3 TJA1145A_M_MC_SLEEP_MODE

```
#define TJA1145A_M_MC_SLEEP_MODE
```

Sleep mode

Definition at line 108 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.4 TJA1145A_M_MC_STANDBY_MODE

```
#define TJA1145A_M_MC_STANDBY_MODE
```

Standby mode

Definition at line 112 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.5 TJA1145A_M_MC_NORMAL_MODE

```
#define TJA1145A_M_MC_NORMAL_MODE
```

Normal mode

Definition at line 116 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.6 TJA1145A_M_NMS_MASK

```
#define TJA1145A_M_NMS_MASK
```

Normal mode status

Definition at line 128 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.7 TJA1145A_M_OTWS_MASK

```
#define TJA1145A_M_OTWS_MASK
```

overtemperature warning status

Definition at line 132 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.8 TJA1145A_M_FSMS_MASK

```
#define TJA1145A_M_FSMS_MASK
```

Sleep mode transition status

Definition at line 136 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.9 TJA1145A_M_NMS_SHIFT

```
#define TJA1145A_M_NMS_SHIFT
```

Normal mode status

Definition at line 141 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.10 TJA1145A_M_OTWS_SHIFT

```
#define TJA1145A_M_OTWS_SHIFT
```

overtemperature warning status

Definition at line 145 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.11 TJA1145A_M_FSMS_SHIFT

```
#define TJA1145A_M_FSMS_SHIFT
```

Sleep mode transition status

Definition at line 149 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.12 TJA1145A_M_NMS_WAS_OR_IS_IN_NORMAL_MODE

```
#define TJA1145A_M_NMS_WAS_OR_IS_IN_NORMAL_MODE
```

TJA1145AA has entered Normal mode (after power-up)

Definition at line 154 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.13 TJA1145A_M_NMS_HAS_NOT_BEEN_IN_NORMAL_MODE

```
#define TJA1145A_M_NMS_HAS_NOT_BEEN_IN_NORMAL_MODE
```

TJA1145AA has not yet been in Normal mode since recent power-up

Definition at line 158 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.14 TJA1145A_M_OTWS_TEMPERATURE_BELOW_WARNING_THRESHOLD

```
#define TJA1145A_M_OTWS_TEMPERATURE_BELOW_WARNING_THRESHOLD
```

IC temperature below overtemperature warning threshold

Definition at line 163 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.15 TJA1145A_M_OTWS_TEMPERATURE_ABOVE_WARNING_THRESHOLD

```
#define TJA1145A_M_OTWS_TEMPERATURE_ABOVE_WARNING_THRESHOLD
```

IC temperature above overtemperature warning threshold

Definition at line 167 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.16 TJA1145A_M_FSMS_SLEEP_MODE_REASON_SPI_COMMAND

```
#define TJA1145A_M_FSMS_SLEEP_MODE_REASON_SPI_COMMAND
```

transition to Sleep mode triggered by an SPI command

Definition at line 172 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.17 TJA1145A_M_FSMS_SLEEP_MODE_REASON_UNDERVOLTAGE

```
#define TJA1145A_M_FSMS_SLEEP_MODE_REASON_UNDERVOLTAGE
```

an undervoltage on VCC and/or VIO forced a transition to Sleep mode

Definition at line 176 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.18 TJA1145A_M_SPIFE_MASK

```
#define TJA1145A_M_SPIFE_MASK
```

SPI failure enable

Definition at line 188 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.19 TJA1145A_M_OTWE_MASK

```
#define TJA1145A_M_OTWE_MASK
```

overtemperature warning enable

Definition at line 192 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.20 TJA1145A_M_SPIFE_SHIFT

```
#define TJA1145A_M_SPIFE_SHIFT
```

SPI failure enable

Definition at line 197 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.21 TJA1145A_M_OTWE_SHIFT

```
#define TJA1145A_M_OTWE_SHIFT
```

overtemperature warning enable

Definition at line 201 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.22 TJA1145A_M_SPIFE_DISABLED

```
#define TJA1145A_M_SPIFE_DISABLED
```

SPI failure detection disabled

Definition at line 206 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.23 TJA1145A_M_SPIFE_ENABLED

```
#define TJA1145A_M_SPIFE_ENABLED
```

SPI failure detection enabled

Definition at line 210 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.24 TJA1145A_M_OTWE_DISABLED

```
#define TJA1145A_M_OTWE_DISABLED
```

overtemperature warning disabled

Definition at line 215 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.25 TJA1145A_M_OTWE_ENABLED

```
#define TJA1145A_M_OTWE_ENABLED
```

overtemperature warning enabled

Definition at line 219 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.26 TJA1145A_M_GPM_MASK [1/4]

```
#define TJA1145A_M_GPM_MASK
```

General purpose memory

Definition at line 282 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.27 TJA1145A_M_GPM_MASK [2/4]

```
#define TJA1145A_M_GPM_MASK
```

General purpose memory

Definition at line 282 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.28 TJA1145A_M_GPM_MASK [3/4]

```
#define TJA1145A_M_GPM_MASK
```

General purpose memory

Definition at line 282 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.29 TJA1145A_M_GPM_MASK [4/4]

```
#define TJA1145A_M_GPM_MASK
```

General purpose memory

Definition at line 282 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.30 TJA1145A_M_GPM_SHIFT [1/4]

```
#define TJA1145A_M_GPM_SHIFT
```

General purpose memory

Definition at line 287 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.31 TJA1145A_M_GPM_SHIFT [2/4]

```
#define TJA1145A_M_GPM_SHIFT
```

General purpose memory

Definition at line 287 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.32 TJA1145A_M_GPM_SHIFT [3/4]

```
#define TJA1145A_M_GPM_SHIFT
```

General purpose memory

Definition at line 287 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.33 TJA1145A_M_GPM_SHIFT [4/4]

```
#define TJA1145A_M_GPM_SHIFT
```

General purpose memory

Definition at line 287 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.34 TJA1145A_M_LK0C_MASK

```
#define TJA1145A_M_LK0C_MASK
```

lock control 0: address area 0x06 to 0x09 - general-purpose memory

Definition at line 299 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.35 TJA1145A_M_LK1C_MASK

```
#define TJA1145A_M_LK1C_MASK
```

lock control 1: address area 0x10 to 0x1F

Definition at line 303 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.36 TJA1145A_M_LK2C_MASK

```
#define TJA1145A_M_LK2C_MASK
```

lock control 2: address area 0x20 to 0x2F - transceiver control and partial networking

Definition at line 307 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.37 TJA1145A_M_LK3C_MASK

```
#define TJA1145A_M_LK3C_MASK
```

lock control 3: address area 0x30 to 0x3F

Definition at line 311 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.38 TJA1145A_M_LK4C_MASK

```
#define TJA1145A_M_LK4C_MASK
```

lock control 4: address area 0x40 to 0x4F - WAKE pin control

Definition at line 315 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.39 TJA1145A_M_LK5C_MASK

```
#define TJA1145A_M_LK5C_MASK
```

lock control 5: address area 0x50 to 0x5F

Definition at line 319 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.40 TJA1145A_M_LK6C_MASK

```
#define TJA1145A_M_LK6C_MASK
```

lock control 6: address area 0x68 to 0x6F

Definition at line 323 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.41 TJA1145A_M_LK0C_SHIFT

```
#define TJA1145A_M_LK0C_SHIFT
```

lock control 0: address area 0x06 to 0x09 - general-purpose memory

Definition at line 328 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.42 TJA1145A_M_LK1C_SHIFT

```
#define TJA1145A_M_LK1C_SHIFT
```

lock control 1: address area 0x10 to 0x1F

Definition at line 332 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.43 TJA1145A_M_LK2C_SHIFT

```
#define TJA1145A_M_LK2C_SHIFT
```

lock control 2: address area 0x20 to 0x2F - transceiver control and partial networking

Definition at line 336 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.44 TJA1145A_M_LK3C_SHIFT

```
#define TJA1145A_M_LK3C_SHIFT
```

lock control 3: address area 0x30 to 0x3F

Definition at line 340 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.45 TJA1145A_M_LK4C_SHIFT

```
#define TJA1145A_M_LK4C_SHIFT
```

lock control 4: address area 0x40 to 0x4F - WAKE pin control

Definition at line 344 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.46 TJA1145A_M_LK5C_SHIFT

```
#define TJA1145A_M_LK5C_SHIFT
```

lock control 5: address area 0x50 to 0x5F

Definition at line 348 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.47 TJA1145A_M_LK6C_SHIFT

```
#define TJA1145A_M_LK6C_SHIFT
```

lock control 6: address area 0x68 to 0x6F

Definition at line 352 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.48 TJA1145A_M_LK0C_WRITING_ENABLED

```
#define TJA1145A_M_LK0C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 357 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.49 TJA1145A_M_LK0C_WRITING_DISABLED

```
#define TJA1145A_M_LK0C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 361 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.50 TJA1145A_M_LK1C_WRITING_ENABLED

```
#define TJA1145A_M_LK1C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 366 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.51 TJA1145A_M_LK1C_WRITING_DISABLED

```
#define TJA1145A_M_LK1C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 370 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.52 TJA1145A_M_LK2C_WRITING_ENABLED

```
#define TJA1145A_M_LK2C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 375 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.53 TJA1145A_M_LK2C_WRITING_DISABLED

```
#define TJA1145A_M_LK2C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 379 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.54 TJA1145A_M_LK3C_WRITING_ENABLED

```
#define TJA1145A_M_LK3C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 384 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.55 TJA1145A_M_LK3C_WRITING_DISABLED

```
#define TJA1145A_M_LK3C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 388 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.56 TJA1145A_M_LK4C_WRITING_ENABLED

```
#define TJA1145A_M_LK4C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 393 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.57 TJA1145A_M_LK4C_WRITING_DISABLED

```
#define TJA1145A_M_LK4C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 397 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.58 TJA1145A_M_LK5C_WRITING_ENABLED

```
#define TJA1145A_M_LK5C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 402 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.59 TJA1145A_M_LK5C_WRITING_DISABLED

```
#define TJA1145A_M_LK5C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 406 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.60 TJA1145A_M_LK6C_WRITING_ENABLED

```
#define TJA1145A_M_LK6C_WRITING_ENABLED
```

SPI write access enabled

Definition at line 411 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.61 TJA1145A_M_LK6C_WRITING_DISABLED

```
#define TJA1145A_M_LK6C_WRITING_DISABLED
```

SPI write access disabled

Definition at line 415 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.62 TJA1145A_FS_I_CMC_MASK

```
#define TJA1145A_FS_I_CMC_MASK
```

CAN transceiver operating mode selection

Definition at line 427 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.63 TJA1145A_FS_I_CPNC_MASK

```
#define TJA1145A_FS_I_CPNC_MASK
```

CAN selective wake-up; when enabled, node is part of a partial network

Definition at line 431 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.64 TJA1145A_FS_I_PNCOK_MASK

```
#define TJA1145A_FS_I_PNCOK_MASK
```

CAN partial networking configuration

Definition at line 435 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.65 TJA1145A_FS_I_CFDC_MASK

```
#define TJA1145A_FS_I_CFDC_MASK
```

CAN FD control(TJA1145AATK/FD only; otherwise ignored)

Definition at line 439 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.66 TJA1145A_FS_I_CMC_SHIFT

```
#define TJA1145A_FS_I_CMC_SHIFT
```

CAN transceiver operating mode selection

Definition at line 444 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.67 TJA1145A_FS_I_CPNC_SHIFT

```
#define TJA1145A_FS_I_CPNC_SHIFT
```

CAN selective wake-up; when enabled, node is part of a partial network

Definition at line 448 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.68 TJA1145A_FS_I_PNCOK_SHIFT

```
#define TJA1145A_FS_I_PNCOK_SHIFT
```

CAN partial networking configuration

Definition at line 452 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.69 TJA1145A_FS_I_CFDC_SHIFT

```
#define TJA1145A_FS_I_CFDC_SHIFT
```

CAN FD control(TJA1145AATK/FD only; otherwise ignored)

Definition at line 456 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.70 TJA1145A_FS_I_CMC_OFFLINE_MODE

```
#define TJA1145A_FS_I_CMC_OFFLINE_MODE
```

Offline mode

Definition at line 461 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.71 TJA1145A_FS_I_CMC_ACTIVE_MODE__01

```
#define TJA1145A_FS_I_CMC_ACTIVE_MODE__01
```

Active mode (when the TJA1145AA is in Normal mode); Vcc undervoltage detection active; transition to Active mode, and remaining in Active mode, requires Vcc above undervoltage threshold

Definition at line 466 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.72 TJA1145A_FS_I_CMC_ACTIVE_MODE__10

```
#define TJA1145A_FS_I_CMC_ACTIVE_MODE__10
```

Active mode (while TJA1145AA is in Normal mode); Vcc undervoltage detection inactive; transition to Active mode requires Vcc above undervoltage threshold

Definition at line 471 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.73 TJA1145A_FS_I_CMC_LISTEN_ONLY_MODE

```
#define TJA1145A_FS_I_CMC_LISTEN_ONLY_MODE
```

Listen-only mode

Definition at line 475 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.74 TJA1145A_FS_I_CPNC_SELECTIVE_WAKE_DISABLED

```
#define TJA1145A_FS_I_CPNC_SELECTIVE_WAKE_DISABLED
```

disable CAN selective wake-up

Definition at line 480 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.75 TJA1145A_FS_I_CPNC_SELECTIVE_WAKE_ENABLED

```
#define TJA1145A_FS_I_CPNC_SELECTIVE_WAKE_ENABLED
```

enable CAN selective wake-up

Definition at line 484 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.76 TJA1145A_FS_I_PNCOK_PN_CONFIGURED

```
#define TJA1145A_FS_I_PNCOK_PN_CONFIGURED
```

partial networking registers configured successfully

Definition at line 489 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.77 TJA1145A_FS_I_CFDC_FD_DISABLED

```
#define TJA1145A_FS_I_CFDC_FD_DISABLED
```

CAN FD tolerance disabled

Definition at line 494 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.78 TJA1145A_FS_I_CFDC_FD_ENABLED

```
#define TJA1145A_FS_I_CFDC_FD_ENABLED
```

CAN FD tolerance enabled

Definition at line 498 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.79 TJA1145A_FS_I_CFS_MASK

```
#define TJA1145A_FS_I_CFS_MASK
```

CAN failure status

Definition at line 510 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.80 TJA1145A_FS_I_VCS_MASK

```
#define TJA1145A_FS_I_VCS_MASK
```

Vcc supply voltage status

Definition at line 514 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.81 TJA1145A_FS_I_CBSS_MASK

```
#define TJA1145A_FS_I_CBSS_MASK
```

CAN-bus silence status

Definition at line 518 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.82 TJA1145A_FS_I_COSCS_MASK

```
#define TJA1145A_FS_I_COSCS_MASK
```

CAN oscillator status

Definition at line 522 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.83 TJA1145A_FS_I_CPNS_MASK

```
#define TJA1145A_FS_I_CPNS_MASK
```

CAN partial networking status

Definition at line 526 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.84 TJA1145A_FS_I_CPNERR_MASK

```
#define TJA1145A_FS_I_CPNERR_MASK
```

CAN partial networking error status

Definition at line 530 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.85 TJA1145A_FS_I_CTS_MASK

```
#define TJA1145A_FS_I_CTS_MASK
```

CAN transceiver status

Definition at line 534 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.86 TJA1145A_FS_I_CFS_SHIFT

```
#define TJA1145A_FS_I_CFS_SHIFT
```

CAN failure status

Definition at line 539 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.87 TJA1145A_FS_I_VCS_SHIFT

```
#define TJA1145A_FS_I_VCS_SHIFT
```

Vcc supply voltage status

Definition at line 543 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.88 TJA1145A_FS_I_CBSS_SHIFT

```
#define TJA1145A_FS_I_CBSS_SHIFT
```

CAN-bus silence status

Definition at line 547 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.89 TJA1145A_FS_I_COSCS_SHIFT

```
#define TJA1145A_FS_I_COSCS_SHIFT
```

CAN oscillator status

Definition at line 551 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.90 TJA1145A_FS_I_CPNS_SHIFT

```
#define TJA1145A_FS_I_CPNS_SHIFT
```

CAN partial networking status

Definition at line 555 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.91 TJA1145A_FS_I_CPNERR_SHIFT

```
#define TJA1145A_FS_I_CPNERR_SHIFT
```

CAN partial networking error status

Definition at line 559 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.92 TJA1145A_FS_I_CTS_SHIFT

```
#define TJA1145A_FS_I_CTS_SHIFT
```

CAN transceiver status

Definition at line 563 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.93 TJA1145A_FS_I_CFS_NO_TIME_OUT_EVENT_DETECTED

```
#define TJA1145A_FS_I_CFS_NO_TIME_OUT_EVENT_DETECTED
```

no TXD dominant time-out event detected

Definition at line 568 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.94 TJA1145A_FS_I_CFS_TIME_OUT_EVENT_DETECTED

```
#define TJA1145A_FS_I_CFS_TIME_OUT_EVENT_DETECTED
```

CAN transmitter disabled due to a TXD dominant time-out event

Definition at line 572 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.95 TJA1145A_FS_I_VCS_ABOVE_UNDERVOLTAGE_THRESHOLD

```
#define TJA1145A_FS_I_VCS_ABOVE_UNDERVOLTAGE_THRESHOLD
```

Vcc is above undervoltage detection threshold

Definition at line 577 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.96 TJA1145A_FS_I_VCS_BELOW_UNDERVOLTAGE_THRESHOLD

```
#define TJA1145A_FS_I_VCS_BELOW_UNDERVOLTAGE_THRESHOLD
```

Vcc is below undervoltage detection threshold

Definition at line 581 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.97 TJA1145A_FS_I_CBSS_CAN_BUS_ACTIVE

```
#define TJA1145A_FS_I_CBSS_CAN_BUS_ACTIVE
```

CAN-bus active (communication detected on bus)

Definition at line 586 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.98 TJA1145A_FS_I_CBSS_CAN_BUS_INACTIVE

```
#define TJA1145A_FS_I_CBSS_CAN_BUS_INACTIVE
```

CAN-bus inactive (for longer than tto(silence))

Definition at line 590 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.99 TJA1145A_FS_I_COSCS_NOT_AT_TARGET_FREQUENCY

```
#define TJA1145A_FS_I_COSCS_NOT_AT_TARGET_FREQUENCY
```

CAN partial networking oscillator unning at target frequency

Definition at line 595 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.100 TJA1145A_FS_I_COSCS_AT_TARGET_FREQUENCY

```
#define TJA1145A_FS_I_COSCS_AT_TARGET_FREQUENCY
```

CAN partial networking oscillator running at target frequency

Definition at line 599 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.101 TJA1145A_FS_I_CPNS_CONFIGURATION_ERROR_DETECTED

```
#define TJA1145A_FS_I_CPNS_CONFIGURATION_ERROR_DETECTED
```

CAN partial networking configuration error detected (PNCOK = 0)

Definition at line 604 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.102 TJA1145A_FS_I_CPNS_CONFIGURATION_OK

```
#define TJA1145A_FS_I_CPNS_CONFIGURATION_OK
```

CAN partial networking configuration OK (PNCOK = 1)

Definition at line 608 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.103 TJA1145A_FS_I_CPNERR_NO_ERROR_DETECTED

```
#define TJA1145A_FS_I_CPNERR_NO_ERROR_DETECTED
```

no CAN partial networking error detected (PNFDE = 0 AND PNCOK = 1)

Definition at line 613 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.104 TJA1145A_FS_I_CPNERR_ERROR_DETECTED

```
#define TJA1145A_FS_I_CPNERR_ERROR_DETECTED
```

CAN partial networking error detected (PNFDE = 1 OR PNCOK = 0; wake-up via standard wake-up pattern only)

Definition at line 617 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.105 TJA1145A_FS_I_CTS_NOT_IN_ACTIVE_MODE

```
#define TJA1145A_FS_I_CTS_NOT_IN_ACTIVE_MODE
```

CAN transceiver not in Active mode

Definition at line 622 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.106 TJA1145A_FS_I_CTS_IN_ACTIVE_MODE

```
#define TJA1145A_FS_I_CTS_IN_ACTIVE_MODE
```

CAN transceiver in Active mode

Definition at line 626 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.107 TJA1145A_FS_I_CWE_MASK

```
#define TJA1145A_FS_I_CWE_MASK
```

CAN wake-up enable

Definition at line 638 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.108 TJA1145A_FS_I_CFE_MASK

```
#define TJA1145A_FS_I_CFE_MASK
```

CAN failure enable

Definition at line 642 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.109 TJA1145A_FS_I_CBSE_MASK

```
#define TJA1145A_FS_I_CBSE_MASK
```

CAN-bus silence enable

Definition at line 646 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.110 TJA1145A_FS_I_CWE_SHIFT

```
#define TJA1145A_FS_I_CWE_SHIFT
```

CAN wake-up enable

Definition at line 651 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.111 TJA1145A_FS_I_CFE_SHIFT

```
#define TJA1145A_FS_I_CFE_SHIFT
```

CAN failure enable

Definition at line 655 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.112 TJA1145A_FS_I_CBSE_SHIFT

```
#define TJA1145A_FS_I_CBSE_SHIFT
```

CAN-bus silence enable

Definition at line 659 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.113 TJA1145A_FS_I_CWE_DISABLED

```
#define TJA1145A_FS_I_CWE_DISABLED
```

CAN wake-up detection disabled

Definition at line 664 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.114 TJA1145A_FS_I_CWE_ENABLED

```
#define TJA1145A_FS_I_CWE_ENABLED
```

CAN wake-up detection enabled

Definition at line 668 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.115 TJA1145A_FS_I_CFE_DISABLED

```
#define TJA1145A_FS_I_CFE_DISABLED
```

CAN failure detection disabled

Definition at line 673 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.116 TJA1145A_FS_I_CFE_ENABLED

```
#define TJA1145A_FS_I_CFE_ENABLED
```

CAN failure detection enabled

Definition at line 677 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.117 TJA1145A_FS_I_CBSE_DISABLED

```
#define TJA1145A_FS_I_CBSE_DISABLED
```

CAN-bus silence detection disabled

Definition at line 682 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.118 TJA1145A_FS_I_CBSE_ENABLED

```
#define TJA1145A_FS_I_CBSE_ENABLED
```

CAN-bus silence detection enabled

Definition at line 686 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.119 TJA1145A_FS_I_CDR_MASK

```
#define TJA1145A_FS_I_CDR_MASK
```

CAN data rate selection

Definition at line 698 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.120 TJA1145A_FS_I_CDR_SHIFT

```
#define TJA1145A_FS_I_CDR_SHIFT
```

CAN data rate selection

Definition at line 703 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.121 TJA1145A_FS_I_CDR_50_KBIT_S

```
#define TJA1145A_FS_I_CDR_50_KBIT_S
```

50 kbit/s

Definition at line 708 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.122 TJA1145A_FS_I_CDR_100_KBIT_S

```
#define TJA1145A_FS_I_CDR_100_KBIT_S
```

100 kbit/s

Definition at line 712 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.123 TJA1145A_FS_I_CDR_125_KBIT_S

```
#define TJA1145A_FS_I_CDR_125_KBIT_S
```

125 kbit/s

Definition at line 716 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.124 TJA1145A_FS_I_CDR_250_KBIT_S

```
#define TJA1145A_FS_I_CDR_250_KBIT_S
```

250 kbit/s

Definition at line 720 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.125 TJA1145A_FS_I_CDR_500_KBIT_S

```
#define TJA1145A_FS_I_CDR_500_KBIT_S
```

500 kbit/s

Definition at line 724 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.126 TJA1145A_FS_I_CDR_1000_KBIT_S

```
#define TJA1145A_FS_I_CDR_1000_KBIT_S
```

1000 kbit/s

Definition at line 728 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.127 TJA1145A_FS_I_ID0_MASK

```
#define TJA1145A_FS_I_ID0_MASK
```

bits ID07 to ID00 of the extended frame format

Definition at line 740 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.128 TJA1145A_FS_I_ID0_SHIFT

```
#define TJA1145A_FS_I_ID0_SHIFT
```

bits ID07 to ID00 of the extended frame format

Definition at line 745 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.129 TJA1145A_FS_I_M_MASK

```
#define TJA1145A_FS_I_M_MASK
```

ID mask bits 7 to 0 of the extended frame format

Definition at line 778 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.130 TJA1145A_FS_I_M_SHIFT

```
#define TJA1145A_FS_I_M_SHIFT
```

ID mask bits 7 to 0 of the extended frame format

Definition at line 783 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.131 TJA1145A_FS_I_DLC_MASK

```
#define TJA1145A_FS_I_DLC_MASK
```

number of data bytes expected in a CAN frame

Definition at line 816 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.132 TJA1145A_FS_I_PNDM_MASK

```
#define TJA1145A_FS_I_PNDM_MASK
```

partial networking data mask

Definition at line 820 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.133 TJA1145A_FS_I_IDE_MASK

```
#define TJA1145A_FS_I_IDE_MASK
```

identifier format

Definition at line 824 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.134 TJA1145A_FS_I_DLC_SHIFT

```
#define TJA1145A_FS_I_DLC_SHIFT
```

number of data bytes expected in a CAN frame

Definition at line 829 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.135 TJA1145A_FS_I_PNDM_SHIFT

```
#define TJA1145A_FS_I_PNDM_SHIFT
```

partial networking data mask

Definition at line 833 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.136 TJA1145A_FS_I_IDE_SHIFT

```
#define TJA1145A_FS_I_IDE_SHIFT
```

identifier format

Definition at line 837 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.137 TJA1145A_FS_I_DLC_0

```
#define TJA1145A_FS_I_DLC_0
```

0

Definition at line 842 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.138 TJA1145A_FS_I_DLC_1

```
#define TJA1145A_FS_I_DLC_1
```

1

Definition at line 846 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.139 TJA1145A_FS_I_DLC_2

```
#define TJA1145A_FS_I_DLC_2
```

2

Definition at line 850 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.140 TJA1145A_FS_I_DLC_3

```
#define TJA1145A_FS_I_DLC_3
```

3

Definition at line 854 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.141 TJA1145A_FS_I_DLC_4

```
#define TJA1145A_FS_I_DLC_4
```

4

Definition at line 858 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.142 TJA1145A_FS_I_DLC_5

```
#define TJA1145A_FS_I_DLC_5
```

```
5
```

Definition at line 862 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.143 TJA1145A_FS_I_DLC_6

```
#define TJA1145A_FS_I_DLC_6
```

```
6
```

Definition at line 866 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.144 TJA1145A_FS_I_DLC_7

```
#define TJA1145A_FS_I_DLC_7
```

```
7
```

Definition at line 870 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.145 TJA1145A_FS_I_DLC_8

```
#define TJA1145A_FS_I_DLC_8
```

```
8
```

Definition at line 874 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.146 TJA1145A_FS_I_DLC_1001__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1001__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 878 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.147 TJA1145A_FS_I_DLC_1010__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1010__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 882 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.148 TJA1145A_FS_I_DLC_1011__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1011__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 886 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.149 TJA1145A_FS_I_DLC_1100__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1100__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 890 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.150 TJA1145A_FS_I_DLC_1101__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1101__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 894 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.151 TJA1145A_FS_I_DLC_1110__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1110__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 898 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.152 TJA1145A_FS_I_DLC_1111__8_BYTES_EXPECTED__

```
#define TJA1145A_FS_I_DLC_1111__8_BYTES_EXPECTED__
```

tolerated, 8 bytes expected

Definition at line 902 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.153 TJA1145A_FS_I_PNDM_DLC____DATA__DON_T_CARE__

```
#define TJA1145A_FS_I_PNDM_DLC____DATA__DON_T_CARE__
```

data length code and data field are *don_t care* for wake-up

Definition at line 907 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.154 TJA1145A_FS_I_PNDM_DLC____DATA_EVALUATED

```
#define TJA1145A_FS_I_PNDM_DLC____DATA_EVALUATED
```

data length code and data field are evaluated at wake-up

Definition at line 911 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.155 TJA1145A_FS_I_IDE_STANDARD__11_BIT__

```
#define TJA1145A_FS_I_IDE_STANDARD__11_BIT__
```

standard frame format (11-bit)

Definition at line 916 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.156 TJA1145A_FS_I_IDE_EXTENDED__29_BIT__

```
#define TJA1145A_FS_I_IDE_EXTENDED__29_BIT__
```

extended frame format (29-bit)

Definition at line 920 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.157 TJA1145A_FS_I_DM0_MASK

```
#define TJA1145A_FS_I_DM0_MASK
```

data mask 0 configuration

Definition at line 932 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.158 TJA1145A_FS_I_DM0_SHIFT

```
#define TJA1145A_FS_I_DM0_SHIFT
```

data mask 0 configuration

Definition at line 937 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.159 TJA1145A_FS_I_DM1_MASK

```
#define TJA1145A_FS_I_DM1_MASK
```

data mask 1 configuration

Definition at line 949 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.160 TJA1145A_FS_I_DM1_SHIFT

```
#define TJA1145A_FS_I_DM1_SHIFT
```

data mask 1 configuration

Definition at line 954 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.161 TJA1145A_FS_I_DM2_MASK

```
#define TJA1145A_FS_I_DM2_MASK
```

data mask 2 configuration

Definition at line 966 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.162 TJA1145A_FS_I_DM2_SHIFT

```
#define TJA1145A_FS_I_DM2_SHIFT
```

data mask 2 configuration

Definition at line 971 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.163 TJA1145A_FS_I_DM3_MASK

```
#define TJA1145A_FS_I_DM3_MASK
```

data mask 3 configuration

Definition at line 983 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.164 TJA1145A_FS_I_DM3_SHIFT

```
#define TJA1145A_FS_I_DM3_SHIFT
```

data mask 3 configuration

Definition at line 988 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.165 TJA1145A_FS_I_DM4_MASK

```
#define TJA1145A_FS_I_DM4_MASK
```

data mask 4 configuration

Definition at line 1000 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.166 TJA1145A_FS_I_DM4_SHIFT

```
#define TJA1145A_FS_I_DM4_SHIFT
```

data mask 4 configuration

Definition at line 1005 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.167 TJA1145A_FS_I_DM5_MASK

```
#define TJA1145A_FS_I_DM5_MASK
```

data mask 5 configuration

Definition at line 1017 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.168 TJA1145A_FS_I_DM5_SHIFT

```
#define TJA1145A_FS_I_DM5_SHIFT
```

data mask 5 configuration

Definition at line 1022 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.169 TJA1145A_FS_I_DM6_MASK

```
#define TJA1145A_FS_I_DM6_MASK
```

data mask 6 configuration

Definition at line 1034 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.170 TJA1145A_FS_I_DM6_SHIFT

```
#define TJA1145A_FS_I_DM6_SHIFT
```

data mask 6 configuration

Definition at line 1039 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.171 TJA1145A_FS_I_DM7_MASK

```
#define TJA1145A_FS_I_DM7_MASK
```

data mask 7 configuration

Definition at line 1051 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.172 TJA1145A_FS_I_DM7_SHIFT

```
#define TJA1145A_FS_I_DM7_SHIFT
```

data mask 7 configuration

Definition at line 1056 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.173 TJA1145A_FS_WPVS_MASK

```
#define TJA1145A_FS_WPVS_MASK
```

WAKE pin status

Definition at line 1068 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.174 TJA1145A_FS_WPVS_SHIFT

```
#define TJA1145A_FS_WPVS_SHIFT
```

WAKE pin status

Definition at line 1073 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.175 TJA1145A_FS_WPVS_BELOW_SWITCHING_THRESHOLD

```
#define TJA1145A_FS_WPVS_BELOW_SWITCHING_THRESHOLD
```

voltage on WAKE pin below switching threshold ($V_{th}(sw)$)

Definition at line 1078 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.176 TJA1145A_FS_WPVS_ABOVE_SWITCHING_THRESHOLD

```
#define TJA1145A_FS_WPVS_ABOVE_SWITCHING_THRESHOLD
```

voltage on WAKE pin above switching threshold ($V_{th}(sw)$)

Definition at line 1082 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.177 TJA1145A_FS_WPFE_MASK

```
#define TJA1145A_FS_WPFE_MASK
```

WAKE pin falling-edge enable

Definition at line 1094 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.178 TJA1145A_FS_WPRE_MASK

```
#define TJA1145A_FS_WPRE_MASK
```

WAKE pin rising-edge enable

Definition at line 1098 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.179 TJA1145A_FS_WPFE_SHIFT

```
#define TJA1145A_FS_WPFE_SHIFT
```

WAKE pin falling-edge enable

Definition at line 1103 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.180 TJA1145A_FS_WPRE_SHIFT

```
#define TJA1145A_FS_WPRE_SHIFT
```

WAKE pin rising-edge enable

Definition at line 1107 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.181 TJA1145A_FS_WPFE_FALLING_EDGE_DISABLED

```
#define TJA1145A_FS_WPFE_FALLING_EDGE_DISABLED
```

falling-edge detection on WAKE pin disabled

Definition at line 1112 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.182 TJA1145A_FS_WPFE_FALLING_EDGE_ENABLED

```
#define TJA1145A_FS_WPFE_FALLING_EDGE_ENABLED
```

falling-edge detection on WAKE pin enabled

Definition at line 1116 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.183 TJA1145A_FS_WPRE_RISING_EDGE_DISABLED

```
#define TJA1145A_FS_WPRE_RISING_EDGE_DISABLED
```

rising-edge detection on WAKE pin disabled

Definition at line 1121 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.184 TJA1145A_FS_WPRE_RISING_EDGE_ENABLED

```
#define TJA1145A_FS_WPRE_RISING_EDGE_ENABLED
```

rising-edge detection on WAKE pin enabled

Definition at line 1125 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.185 TJA1145A_FS_SYSE_MASK

```
#define TJA1145A_FS_SYSE_MASK
```

system event

Definition at line 1137 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.186 TJA1145A_FS_TRXE_MASK

```
#define TJA1145A_FS_TRXE_MASK
```

transceiver event

Definition at line 1141 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.187 TJA1145A_FS_WPE_MASK

```
#define TJA1145A_FS_WPE_MASK
```

WAKE pin event

Definition at line 1145 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.188 TJA1145A_FS_SYSE_SHIFT

```
#define TJA1145A_FS_SYSE_SHIFT
```

system event

Definition at line 1150 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.189 TJA1145A_FS_TRXE_SHIFT

```
#define TJA1145A_FS_TRXE_SHIFT
```

transceiver event

Definition at line 1154 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.190 TJA1145A_FS_WPE_SHIFT

```
#define TJA1145A_FS_WPE_SHIFT
```

WAKE pin event

Definition at line 1158 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.191 TJA1145A_FS_SYSE_NO_PENDING_EVENTS

```
#define TJA1145A_FS_SYSE_NO_PENDING_EVENTS
```

no pending system event

Definition at line 1163 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.192 TJA1145A_FS_SYSE_PENDING_EVENT_S_

```
#define TJA1145A_FS_SYSE_PENDING_EVENT_S_
```

system event pending at address 0x61

Definition at line 1167 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.193 TJA1145A_FS_TRXE_NO_PENDING_EVENTS

```
#define TJA1145A_FS_TRXE_NO_PENDING_EVENTS
```

no pending transceiver event

Definition at line 1172 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.194 TJA1145A_FS_TRXE_PENDING_EVENT_S_

```
#define TJA1145A_FS_TRXE_PENDING_EVENT_S_
```

transceiver event pending at address 0x63

Definition at line 1176 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.195 TJA1145A_FS_WPE_NO_PENDING_EVENTS

```
#define TJA1145A_FS_WPE_NO_PENDING_EVENTS
```

no pending WAKE pin event

Definition at line 1181 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.196 TJA1145A_FS_WPE_PENDING_EVENT_S_

```
#define TJA1145A_FS_WPE_PENDING_EVENT_S_
```

WAKE pin event pending at address 0x64

Definition at line 1185 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.197 TJA1145A_FS_SPIF_MASK

```
#define TJA1145A_FS_SPIF_MASK
```

SPI failure

Definition at line 1197 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.198 TJA1145A_FS_OTW_MASK

```
#define TJA1145A_FS_OTW_MASK
```

overtemperature warning

Definition at line 1201 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.199 TJA1145A_FS_PO_MASK

```
#define TJA1145A_FS_PO_MASK
```

power-on

Definition at line 1205 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.200 TJA1145A_FS_SPIF_SHIFT

```
#define TJA1145A_FS_SPIF_SHIFT
```

SPI failure

Definition at line 1210 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.201 TJA1145A_FS_OTW_SHIFT

```
#define TJA1145A_FS_OTW_SHIFT
```

overtemperature warning

Definition at line 1214 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.202 TJA1145A_FS_PO_SHIFT

```
#define TJA1145A_FS_PO_SHIFT
```

power-on

Definition at line 1218 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.203 TJA1145A_FS_SPIF_SPI_FAILURE_EVENT

```
#define TJA1145A_FS_SPIF_SPI_FAILURE_EVENT
```

SPI failure detected

Definition at line 1223 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.204 TJA1145A_FS_OTW_OVERTEMP_WARNING_EVENT

```
#define TJA1145A_FS_OTW_OVERTEMP_WARNING_EVENT
```

the global chip temperature has exceeded the overtemperature warning threshold, T_th(warn)otp

Definition at line 1228 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.205 TJA1145A_FS_PO_POWER_ON_EVENT

```
#define TJA1145A_FS_PO_POWER_ON_EVENT
```

the TJA1145AA has left Off mode after battery power-on

Definition at line 1233 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.206 TJA1145A_FS_CW_MASK

```
#define TJA1145A_FS_CW_MASK
```

CAN wake-up

Definition at line 1245 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.207 TJA1145A_FS_CF_MASK

```
#define TJA1145A_FS_CF_MASK
```

CAN failure

Definition at line 1249 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.208 TJA1145A_FS_CBS_MASK

```
#define TJA1145A_FS_CBS_MASK
```

CAN-bus status

Definition at line 1253 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.209 TJA1145A_FS_PNFDE_MASK

```
#define TJA1145A_FS_PNFDE_MASK
```

partial networking frame detection error

Definition at line 1257 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.210 TJA1145A_FS_CW_SHIFT

```
#define TJA1145A_FS_CW_SHIFT
```

CAN wake-up

Definition at line 1262 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.211 TJA1145A_FS_CF_SHIFT

```
#define TJA1145A_FS_CF_SHIFT
```

CAN failure

Definition at line 1266 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.212 TJA1145A_FS_CBS_SHIFT

```
#define TJA1145A_FS_CBS_SHIFT
```

CAN-bus status

Definition at line 1270 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.213 TJA1145A_FS_PNFDE_SHIFT

```
#define TJA1145A_FS_PNFDE_SHIFT
```

partial networking frame detection error

Definition at line 1274 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.214 TJA1145A_FS_CW_CAN_WAKE_UP_EVENT

```
#define TJA1145A_FS_CW_CAN_WAKE_UP_EVENT
```

CAN wake-up event detected

Definition at line 1279 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.215 TJA1145A_FS_CF_CAN_FAILURE_EVENT

```
#define TJA1145A_FS_CF_CAN_FAILURE_EVENT
```

CAN failure event detected

Definition at line 1284 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.216 TJA1145A_FS_CBS_BUS_SILENCE_EVENT

```
#define TJA1145A_FS_CBS_BUS_SILENCE_EVENT
```

no activity on CAN-bus for tto(silence)

Definition at line 1289 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.217 TJA1145A_FS_PNFDE_FRAME_ERROR_EVENT

```
#define TJA1145A_FS_PNFDE_FRAME_ERROR_EVENT
```

partial networking frame detection error detected

Definition at line 1294 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.218 TJA1145A_FS_WPF_MASK

```
#define TJA1145A_FS_WPF_MASK
```

WAKE pin falling edge

Definition at line 1306 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.219 TJA1145A_FS_WPR_MASK

```
#define TJA1145A_FS_WPR_MASK
```

WAKE pin rising edge

Definition at line 1310 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.220 TJA1145A_FS_WPF_SHIFT

```
#define TJA1145A_FS_WPF_SHIFT
```

WAKE pin falling edge

Definition at line 1315 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.221 TJA1145A_FS_WPR_SHIFT

```
#define TJA1145A_FS_WPR_SHIFT
```

WAKE pin rising edge

Definition at line 1319 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.222 TJA1145A_FS_WPF_FALLING_EDGE_EVENT

```
#define TJA1145A_FS_WPF_FALLING_EDGE_EVENT
```

falling edge detected on WAKE pin

Definition at line 1324 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.223 TJA1145A_FS_WPR_RISING_EDGE_EVENT

```
#define TJA1145A_FS_WPR_RISING_EDGE_EVENT
```

rising edge detected on WAKE pin

Definition at line 1329 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.224 TJA1145A_FS_IDS_MASK

```
#define TJA1145A_FS_IDS_MASK
```

device identification code

Definition at line 1341 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.225 TJA1145A_FS_IDS_SHIFT

```
#define TJA1145A_FS_IDS_SHIFT
```

device identification code

Definition at line 1346 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.226 TJA1145A_FS_IDS_TJA1145AATK

```
#define TJA1145A_FS_IDS_TJA1145AATK
```

TJA1145AATK

Definition at line 1351 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.3.227 TJA1145A_FS_IDS_TJA1145AATK_FD

```
#define TJA1145A_FS_IDS_TJA1145AATK_FD
```

TJA1145AATK/FD

Definition at line 1355 of file [CanTrcv_tja1145a_Ip_Regs.h](#).

6.1.4 Enum Reference

6.1.4.1 CanTrcv_43_tja1145a_eDriverStatusType

```
enum CanTrcv_43_tja1145a_eDriverStatusType
```

CanTrcv_43_tja1145a_eDriverStatusType.

Driver status helps to prevent double driver initialization.

Enumerator

CANTRCV_43_TJA1145A_UNINIT	Driver not initialized. Driver ready
----------------------------	--------------------------------------

Definition at line 189 of file [CanTrcv_43_tja1145a.h](#).

6.1.4.2 CanTrcv_tja1145a_StatusType

```
enum CanTrcv_tja1145a_StatusType
```

Enumeration of possible results of command execution.

Enumerator

TJA1145A_SUCCESS	No error.
TJA1145A_FAIL	General error, command failed to execute task successfully.
TJA1145A_ERR_SPI	SPI communication error.
TJA1145A_ERR_VALUE	Corrupted Value returned from SPI.
TJA1145A_ERR_PARAM	Wrong parameter.

Definition at line 175 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.3 CanTrcv_tja1145a_SystemErrorType

enum [CanTrcv_tja1145a_SystemErrorType](#)

System error.

Enumerator

TJA1145A_OVERTEMP	Overtemperature detected.
TJA1145A_PN_ERROR	Partial networking error detected.
TJA1145A_PN_STATUS_ERROR	Partial networking configuration error.
TJA1145A_CAN_OSC_STATUS	CAN oscillator not running at target frequency.
TJA1145A_VCC_UNDERVOLTAGE	VCC undervoltage detected.
TJA1145A_CAN_FAILURE	CAN transmitter disabled due to a TXD dominant time-out event.

Definition at line 193 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.4 CanTrcv_tja1145a_PorEventType

enum [CanTrcv_tja1145a_PorEventType](#)

Power on event flag.

Enumerator

TJA1145A_NO_POR_EVENT	No power on event occurred.
TJA1145A_POR_EVENT_OCCURRED	Power on event occurred.

Definition at line 213 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.5 CanTrcv_tja1145a_WuReasonType

enum [CanTrcv_tja1145a_WuReasonType](#)

Wake up source.

Enumerator

TJA1145A_WU_NO_EVENT	No wake up event occurred.
TJA1145A_WU_PIN_WAKEUP	WAKE pin event.
TJA1145A_WU_CAN_BUS_WAKEUP	CAN bus wake event.

Enumerator

TJA1145A_WU_SPI_FAILURE	SPI failure (triggers wake up only in standby mode).
TJA1145A_WU_PN_FRAME_ERROR	Partial networking frame error.
TJA1145A_WU_CAN_BUS_SILENCE	CAN bus wake up frame received.
TJA1145A_WU_INTERNALLY	TJA1145A was put to normal mode via SPI command.

Definition at line 225 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.6 CanTrcv_tja1145a_WuPinModeType

```
enum CanTrcv_tja1145a_WuPinModeType
```

WAKE pin wake up feature mode.

Enumerator

TJA1145A_WU_PIN_DISABLED	WAKE pin functionality disabled.
TJA1145A_WU_PIN_FALLING_EDGE	WAKE pin active on falling edge.
TJA1145A_WU_PIN_RISING_EDGE	WAKE pin active on rising edge.

Definition at line 247 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.7 CanTrcv_tja1145a_WuCanModeType

```
enum CanTrcv_tja1145a_WuCanModeType
```

CAN bus wake up feature mode.

Enumerator

TJA1145A_WU_CAN_BUS_DISABLED	WAKE pin functionality disabled.
TJA1145A_WU_CAN_BUS_ENABLED	WAKE pin functionality disabled.

Definition at line 261 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.8 CanTrcv_tja1145a_CanDataRateType

```
enum CanTrcv_tja1145a_CanDataRateType
```

CAN data rate.

Enumerator

TJA1145A_DR_50KBITS	50kBit/s
TJA1145A_DR_100KBITS	100kBit/s
TJA1145A_DR_125KBITS	125kBit/s
TJA1145A_DR_250KBITS	250kBit/s
TJA1145A_DR_500KBITS	500kBit/s
TJA1145A_DR_1000KBITS	1000kBit/s

Definition at line 273 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.9 CanTrcv_tja1145a_IdFormatType

```
enum CanTrcv_tja1145a_IdFormatType
```

Wake up frame identifier.

Enumerator

TJA1145A_11BIT_IDENTIFIER	Standard frame identifier (11 bit).
TJA1145A_29BIT_IDENTIFIER	Extended frame identifier (29 bit).

Definition at line 293 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.10 CanTrcv_tja1145a_WufEvalType

```
enum CanTrcv_tja1145a_WufEvalType
```

There are 2 options of how a WUF can be evaluated:

1. Only ID of WUF is considered in evaluation.
2. ID, DLC and data bytes are included in WUF evaluation.

Enumerator

TJA1145A_WUF_ID	Only ID is included in evaluation of wake up frame.
TJA1145A_WUF_ID_DLC_DATA	ID, DLC and data is considered in evaluation of wake up frame.

Definition at line 307 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.11 CanTrcv_tja1145a_CanControllerModeType

enum [CanTrcv_tja1145a_CanControllerModeType](#)

CAN controller modes.

Enumerator

TJA1145A_OFFLINE_MODE	Offline mode.
TJA1145A_ACTIVE_MODE_VCC_UV_ACTIVE	CAN controller active mode with VCC undervoltage detection active.
TJA1145A_ACTIVE_MODE_VCC_UV_↔ INACTIVE	CAN controller active mode with VCC undervoltage detection inactive.
TJA1145A_LISTEN_ONLY_MODE	Listen only mode.

Definition at line 319 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.12 CanTrcv_tja1145a_FdModeType

enum [CanTrcv_tja1145a_FdModeType](#)

Flexible data rate (available for TJA1145AT/FD and TJA1145ATK/FD only).

Enumerator

TJA1145A_FD_DISABLED	FD disabled.
TJA1145A_FD_ENABLED	FD enabled.

Definition at line 335 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.13 CanTrcv_tja1145a_VccUvConfigType

enum [CanTrcv_tja1145a_VccUvConfigType](#)

VCC undervoltage detection in CAN active mode configuration. If enabled CAN controller will automatically switch to offline mode if VCC undervoltage is detected.

Enumerator

TJA1145A_VCC_UV_DISABLED	VCC undervoltage detection in active mode disabled.
TJA1145A_VCC_UV_ENABLED	VCC undervoltage detection in active mode enabled.

Definition at line 349 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.14 CanTrcv_tja1145a_PNActivationType

enum [CanTrcv_tja1145a_PNActivationType](#)

Partial networking configuration.

Enumerator

TJA1145A_PN_DISABLED	Partial networking not active.
TJA1145A_PN_ENABLED	Partial networking active.

Definition at line 362 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.15 CanTrcv_tja1145a_TrvcFlagStateType

enum [CanTrcv_tja1145a_TrvcFlagStateType](#)

Indicates flag's state.

Enumerator

TJA1145A_FLAG_SET	Flag set.
TJA1145A_FLAG_CLEARED	Flag cleared.

Definition at line 375 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.4.16 CanTrcv_tja1145a_MemoryRegisterType

enum [CanTrcv_tja1145a_MemoryRegisterType](#)

List of general purpose memory registers.

Enumerator

TJA1145A_MEMORY↔ _0	Memory register 0.
TJA1145A_MEMORY↔ _1	Memory register 1.
TJA1145A_MEMORY↔ _2	Memory register 2.
TJA1145A_MEMORY↔ _3	Memory register 3.

Definition at line 387 of file [CanTrcv_tja1145a_Ip.h](#).

6.1.5 Function Reference

6.1.5.1 CanTrcv_43_tja1145a_Init()

```
void CanTrcv_43_tja1145a_Init (
    const CanTrcv_43_tja1145a_DriverConfigType * ConfigPtr )
```

Initializes CanTrcv module. SID 0x00.

Initializes all transceivers configured in ConfigPtr parameter. The CANTRCV module shall be initialized by [CanTrcv_43_tja1145a_Init\(\)](#) service call during the start-up.

Parameters

in	<i>ConfigPtr</i>	Pointer to driver configuration structure.
----	------------------	--

Returns

void

Precondition

CanTrcv_43_tja1145a_Init shall be called at most once during runtime.

Postcondition

CanTrcv_43_tja1145a_Init shall initialize all the transceivers and set the driver in READY state.

6.1.5.2 CanTrcv_43_tja1145a_SetOpMode()

```
Std_ReturnType CanTrcv_43_tja1145a_SetOpMode (
    uint8 Transceiver,
    CanTrcv_TrsvModeType OpMode )
```

Sets the mode of the Transceiver to the value OpMode. SID 0x01.

Puts the device in one of following modes: normal, standby, sleep.

Parameters

in	<i>Transceiver</i>	Index of the transceiver.
out	<i>OpMode</i>	Desired operating mode.

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_SetOpMode method.

Postcondition

CanTrcv_43_tja1145a_SetOpMode shall set the transceiver in the desired state.

6.1.5.3 CanTrcv_43_tja1145a_GetOpMode()

```
Std_ReturnType CanTrcv_43_tja1145a_GetOpMode (
    uint8 Transceiver,
    CanTrcv_TrsvModeType * OpMode )
```

Gets current operation mode. SID 0x02.

Gets the mode of the Transceiver and returns it in OpMode. The device is in one of following modes: normal, standby, sleep.

Module Documentation

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
out	<i>OpMode</i>	Current operating mode.

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_GetOpMode method.

Postcondition

CanTrcv_43_tja1145a_GetOpMode shall return the currently working mode of the transceiver.

6.1.5.4 CanTrcv_43_tja1145a_GetBusWuReason()

```
Std_ReturnType CanTrcv_43_tja1145a_GetBusWuReason (
    uint8 Transceiver,
    CanTrcv_TrcvWakeupReasonType * Reason )
```

Gets the wakeup reason of the Transceiver and returns it in parameter Reason. SID 0x03.

The device can be woken up by:

- standard wake up sources: WAKE pin, CAN bus or internally
- interrupts: SPI failure, CAN bus silence and PN frame error. Interrupts are collectively reported as CANTRCV_WU_BY_SYSERR.

Parameters

in	<i>Transceiver</i>	CAN transceiver to which API call has to be applied.
out	<i>Reason</i>	Pointer to wake up reason.

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_GetBusWuReason method.

Postcondition

CanTrcv_43_tja1145a_GetBusWuReason shall return wake up reason.

6.1.5.5 CanTrcv_43_tja1145a_GetTrcvSystemData()

```
Std_ReturnType CanTrcv_43_tja1145a_GetTrcvSystemData (
    uint8 Transceiver,
    uint32 * TrcvSysData )
```

Reads transceiver system data. SID 0x09.

Returns following information: transition to normal and sleep mode status, overtemperature flag, CAN controller status, partial networking status, oscillator working properly status, CAN bus silence flag, VCC supply status, CAN failure status.

Parameters

in	<i>Transceiver</i>	CAN transceiver to which API call has to be applied.
out	<i>TrcvSysData</i>	Holds transceiver system status.

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Module Documentation

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_GetTrcvSystemData method.

Postcondition

CanTrcv_43_tja1145a_GetTrcvSystemData shall return status of the transceiver.

6.1.5.6 CanTrcv_43_tja1145a_ClearTrcvWufFlag()

```
Std_ReturnType CanTrcv_43_tja1145a_ClearTrcvWufFlag (
    uint8 Transceiver )
```

Clears the WUF flag in the transceiver hardware. SID 0x0A.

Clears wake up flag and reports this event.

Parameters

in	<i>Transceiver</i>	CAN Transceiver ID.
----	--------------------	---------------------

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_ClearTrcvWufFlag method.

Postcondition

CanTrcv_43_tja1145a_ClearTrcvWufFlag shall return status of the transceiver.

6.1.5.7 CanTrcv_43_tja1145a_SetWakeupMode()

```
Std_ReturnType CanTrcv_43_tja1145a_SetWakeupMode (
    uint8 Transceiver,
    CanTrcv_TrcvWakeupModeType TrcvWakeupMode )
```

Enables, disables or clears wake up events of the Transceiver according to TrcvWakeupMode. SID 0x05.

Enables, disables or clears wake up functionality. If WU mode is disabled all wake up sources and interrupts are off. If WU mode is enabled, all wake up sources and interrupts are set according to configuration. If WU mode is clear, pending wake up flag is cleared.

Parameters

in	<i>Transceiver</i>	CAN Transceiver ID.
in	<i>TrcvWakeupMode</i>	Mode of wake up functionality (enabled, disabled, cleared).

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_SetWakeupMode method.

Postcondition

CanTrcv_43_tja1145a_SetWakeupMode shall return status of the transceiver.

6.1.5.8 CanTrcv_43_tja1145a_CheckWakeup()

```
Std_ReturnType CanTrcv_43_tja1145a_CheckWakeup (
    uint8 Transceiver )
```

Service is called by underlying CANIF in case a wake up interrupt is detected. SID 0x07.

Reads wake up source from the device and reports it to ECUM.

Module Documentation

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
----	--------------------	---------------------

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_CheckWakeup method.

Postcondition

CanTrcv_43_tja1145a_CheckWakeup shall read and report wake up reason.

6.1.5.9 CanTrcv_43_tja1145a_CheckWakeFlag()

```
Std_ReturnType CanTrcv_43_tja1145a_CheckWakeFlag (  
    uint8 Transceiver )
```

Requests to check the status of the wakeup flag from the transceiver hardware. SID 0x0E.

Checks wake up event and if WU occurred, reports it.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
----	--------------------	---------------------

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_CheckWakeFlag method.

Postcondition

CanTrcv_43_tja1145a_CheckWakeFlag shall check for wake up event, and if such event occurred, report it.

6.1.5.10 CanTrcv_43_tja1145a_ReadTrcvTimeoutFlag()

```
Std_ReturnType CanTrcv_43_tja1145a_ReadTrcvTimeoutFlag (
    uint8 Transceiver,
    CanTrcv_tja1145a_TrcvFlagStateType * FlagState )
```

Reads the status of the timeout flag from the transceiver hardware. SID 0x0B.

Reads and returns status of timeout flag.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
out	<i>FlagState</i>	State of the timeout flag.

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Postcondition

CanTrcv_43_tja1145a_ReadTrcvTimeoutFlag shall return timeout flag status.

6.1.5.11 CanTrcv_43_tja1145a_ClearTrcvTimeoutFlag()

```
Std_ReturnType CanTrcv_43_tja1145a_ClearTrcvTimeoutFlag (
    uint8 Transceiver )
```




Module Documentation

Clears the status of the timeout flag in the transceiver hardware. SID 0x0C.

Clears timeout flag.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
----	--------------------	---------------------

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Postcondition

CanTrcv_43_tja1145a_ClearTrcvTimeoutFlag shall clear timeout flag.

6.1.5.12 CanTrcv_43_tja1145a_ReadTrcvSilenceFlag()

```
Std_ReturnType CanTrcv_43_tja1145a_ReadTrcvSilenceFlag (
    uint8 Transceiver,
    CanTrcv_tja1145a_TrvcFlagStateType * FlagState )
```

Reads the status of the silence flag from the transceiver hardware. SID 0x0D.

Reads CAN bus silence flag.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
out	<i>FlagState</i>	State of the silence flag.

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Postcondition

CanTrcv_43_tja1145a_ReadTrcvSilenceFlag shall read CAN bus silence flag.

6.1.5.13 CanTrcv_43_tja1145a_SetPNActivationState()

```
Std_ReturnType CanTrcv_43_tja1145a_SetPNActivationState (
    CanTrcv_tja1145a_PNActivationType ActivationState )
```

Sets PN functionality state. SID 0x0F.

Enables/disables PN feature.

Parameters

in	<i>ActivationState</i>	Enables/disables PN wakeup functionality.
----	------------------------	---

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_SetPNActivationState method.

Postcondition

CanTrcv_43_tja1145a_SetPNActivationState shall set mode of PN feature.

6.1.5.14 CanTrcv_43_tja1145a_Exts_ClearPorFlag()

```
Std_ReturnType CanTrcv_43_tja1145a_Exts_ClearPorFlag (
    uint8 Transceiver )
```

Clears POR flag.

Clears POR flag.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
----	--------------------	---------------------

Returns

Std_ReturnType Result of the transition.

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_Exts_ClearPorFlag method.

Postcondition

CanTrcv_43_tja1145a_Exts_ClearPorFlag shall clear POR flag.

6.1.5.15 CanTrcv_43_tja1145a_Exts_ClearSystemError()

```
Std_ReturnType CanTrcv_43_tja1145a_Exts_ClearSystemError (
    uint8 Transceiver,
    CanTrcv_tja1145a_SystemErrorType SystemError )
```

Clears system error.

Clears following system error flags: overtemperature, PN frame error, CAN bus silence, can failure (Tx dominant timeout). Following flags are read only: PN configuration status (if set, PN must be reconfigured), VCC/VIO undervoltage, oscillator error. This functions fails on an attempt to clear any of these read only flags.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
out	<i>SystemError</i>	Which system error to clear.

Returns

Std_ReturnType Result of the transition

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_Exts_ClearSystemError method.

Postcondition

CanTrcv_43_tja1145a_Exts_ClearSystemError shall clear system error flag.

6.1.5.16 CanTrcv_43_tja1145a_Exts_WriteMemoryRegister()

```
Std_ReturnType CanTrcv_43_tja1145a_Exts_WriteMemoryRegister (
    uint8 Transceiver,
    CanTrcv_tja1145a_MemoryRegisterType MemRegister,
    uint8 Data )
```

Writes general purpose memory register.

Writes one of 4 available general purpose memory registers.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
in	<i>MemRegister</i>	Which memory register to access.
in	<i>Data</i>	Data to write to given memory register.

Returns

Std_ReturnType Result of the transition

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_Exts_WriteMemoryRegister method.

Postcondition

CanTrcv_43_tja1145a_Exts_WriteMemoryRegister shall write data to given memory register.

6.1.5.17 CanTrcv_43_tja1145a_Exts_ReadMemoryRegister()

```
Std_ReturnType CanTrcv_43_tja1145a_Exts_ReadMemoryRegister (
    uint8 Transceiver,
    CanTrcv_tja1145a_MemoryRegisterType MemRegister,
    uint8 * Data )
```

Reads general purpose memory register.

Reads one of 4 available general purpose memory registers.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
in	<i>MemRegister</i>	Which memory register to access.
out	<i>Data</i>	Stores content of given memory register.

Returns

Std_ReturnType Result of the transition

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_Exts_ReadMemoryRegister method.

Postcondition

CanTrcv_43_tja1145a_Exts_ReadMemoryRegister shall read content of given memory register.

6.1.5.18 CanTrcv_43_tja1145a_Exts_SetLockControlRegister()

```
Std_ReturnType CanTrcv_43_tja1145a_Exts_SetLockControlRegister (
    uint8 Transceiver,
    const CanTrcv_tja1145a_LockRegistersType * LockControl )
```

Locks/unlocks register groups.

Each item in the configuration structure locks/unlocks respective register groups. lockRange06h09h - locks/unlocks register within address range 06h - 09h, lockRange10h1Fh - locks/unlocks register within address range 10h - 1Fh, lockRange20h2Fh - locks/unlocks register within address range 20h - 2Fh, lockRange30h3Fh - locks/unlocks register within address range 30h - 3Fh, lockRange40h4Fh - locks/unlocks register within address range 40h - 4Fh, lockRange50h5Fh - locks/unlocks register within address range 50h - 5Fh, lockRange68h6Fh - locks/unlocks register within address range 68h - 6Fh.

Parameters

in	<i>Transceiver</i>	CAN transceiver ID.
in	<i>MemRegister</i>	Which memory register to access.
out	<i>Data</i>	Stores content of given memory register.

Returns

Std_ReturnType Result of the transition

Return values

<i>E_OK</i>	Operation executed successfully.
<i>E_NOT_OK</i>	Operation failed.

Precondition

CanTrcv module should be initialized before calling the CanTrcv_43_tja1145a_Exts_SetLockControlRegister method.

Postcondition

CanTrcv_43_tja1145a_Exts_SetLockControlRegister shall read content of given memory register.

6.1.5.19 CanTrcv_43_tja1145a_MainFunction()

```
void CanTrcv_43_tja1145a_MainFunction (
    void )
```

Service to scan all busses for wake up events and perform these event. SID 0x06.

Scan transceivers for wake up event and reports such event.

Parameters

in	<i>None</i>	
----	-------------	--

Returns

void

Precondition

Postcondition

6.1.5.20 CanTrcv_43_tja1145a_MainFunctionDiagnostics()

```
void CanTrcv_43_tja1145a_MainFunctionDiagnostics (
    void )
```

Reads the transceiver diagnostic status periodically and sets product/development accordingly. SID 0x08.

Reads system error, and reports it if any.

Parameters

in	<i>None</i>	
----	-------------	--

Returns

void

Precondition

Postcondition

6.1.5.21 CanTrcv_43_tja1145a_DeInit()

```
void CanTrcv_43_tja1145a_DeInit (
    void )
```

De-initializes the CanTrcv module. SID 0x10.

De-initialize all the transceivers. The CANTRCV module shall be de-initialized by [CanTrcv_43_tja1145a_DeInit\(\)](#) service call. This routine is called by:

- CanIf or an upper layer according to Autosar requirements.

Parameters

in	None	
----	------	--

Returns

void

Precondition

Before transceiver de-initialization, the driver must be initialized and the transceivers must not be in Start state.

Postcondition

CanTrcv_43_tja1145a_DeInit shall de-initialize all the transceivers and set the driver in UNINIT state.

6.1.5.22 TJA1145A_DRV_Init()

```
void TJA1145A_DRV_Init (
    uint8 Transceiver,
    const CanTrcv_tja1145a_ConfigurationDataType * ConfigData )
```

Initializes the TJA1145A driver. After driver initialization, function TJA1145A_DRV_ConfigureDevice must be called to configure the transceiver and finalize overall configuration. Having driver and device configuration in separate functions enables TJA1145A reconfiguration on cold start (MCU was off) only if POR or error event has been detected.

Parameters

Transceiver	Which device to initialize.
ConfigData	Pointer to configuration of the device and SPI.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.23 TJA1145A_DRV_ConfigureDevice()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ConfigureDevice (
    uint8 Transceiver,
    const CanTrcv_tja1145a_DeviceConfigType * DeviceConfig )
```

Configures the transceiver. Writes data from configuration structure to its registers.

Parameters

<i>Transceiver</i>	Which device to configure.
<i>DeviceConfig</i>	Pointer to TJA1145A configuration.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.24 TJA1145A_DRV_ReadRegister()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadRegister (
    uint8 Transceiver,
    uint8 RegAddress,
    uint8 * RxData )
```

Performs a read from a single TJA1145A register. Reads a single TJA1145A register. Read out data is stored in a location pointed by pRxData.

Parameters

<i>Transceiver</i>	Which device to read from.
<i>RegAddress</i>	Register address.
<i>RxDData</i>	Pointer to memory location where read out data is stored.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.25 TJA1145A_DRV_ReadMultipleRegisters()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadMultipleRegisters (
    uint8 Transceiver,
    uint8 RegAddress,
    uint16 NumRegisters,
    uint8 * RxData )
```

Performs a read from multiple TJA1145A registers. Reads up to three consecutive TJA1145A registers. Address of the first register must be provided via the 'Address' parameter. TJA1145A will then automatically increment addresses of following registers. Read out data is stored in a location pointed by pRxData. Note: it is important to allocate enough space to accommodate all read out data.

Parameters

<i>Transceiver</i>	Which device to read from.
<i>RegAddress</i>	Address of the first register. Addresses of following registers are incremented by TJA1145A.
<i>NumRegisters</i>	Number of registers to read.
<i>RxData</i>	Pointer to memory location where read out data is stored.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.26 TJA1145A_DRV_WriteRegister()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_WriteRegister (
    uint8 Transceiver,
    uint8 RegAddress,
    uint8 TxData )
```

Writes a single TJA1145A register.

Parameters

<i>Transceiver</i>	Device ID.
<i>RegAddress</i>	Register address.
<i>TxData</i>	Data to write.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.27 TJA1145A_DRV_WriteMultipleRegisters()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_WriteMultipleRegisters (
    uint8 Transceiver,
    uint8 RegAddress,
    uint16 NumRegisters,
    const uint8 * TxData )
```

Writes multiple TJA1145A consecutive registers. Writes up to the three consecutive TJA1145A registers. The address of the first register must be provided via the 'Address' parameter. Addresses of following registers are automatically incremented by TJA1145A.

Parameters

<i>Transceiver</i>	Device ID.
<i>RegAddress</i>	Address of the first register.
<i>NumRegisters</i>	Number of registers to write.
<i>TxData</i>	Pointer to memory location with data to write.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.28 TJA1145A_DRV_UpdateRegister()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_UpdateRegister (
    uint8 Transceiver,
    uint8 RegAddress,
    uint8 Value,
    uint8 Mask,
    uint8 Shift )
```

Updates a bit/bit group in a register and leaves the rest of the register unchanged.

Parameters

<i>Transceiver</i>	Device ID.
<i>RegAddress</i>	Address of register to write.
<i>Value</i>	Data to write to register specified by RegAddress parameter.
<i>Mask</i>	Mask of the Value.
<i>Shift</i>	Value position within the register.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.29 TJA1145A_DRV_SetMode()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetMode (
    uint8 Transceiver,
    CanTrcv_TrsvModeType OpMode )
```

Sets the mode of the transceiver to the Value OpMode.

Parameters

<i>Transceiver</i>	Device ID.
<i>OpMode</i>	Desired operational mode.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.30 TJA1145A_DRV_GetMode()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetMode (
    uint8 Transceiver,
    CanTrcv_TrsvModeType * OpMode )
```

Reads the mode of the transceiver from 'Mode control register' and returns it in OpMode parameter.

Parameters

<i>Transceiver</i>	Device ID.
<i>OpMode</i>	Stores the mode of the Transceiver.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.31 TJA1145A_DRV_GetWuReason()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetWuReason (
    uint8 Transceiver,
    CanTrcv_tja1145a_WuReasonType * WuReason )
```

Retrieves the wake up reason. Possible wake up sources are: CAN bus, wake pin (either rising or falling edge).

Parameters

<i>Transceiver</i>	Device ID.
<i>WuReason</i>	Stores the mode of the transceiver.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.32 TJA1145A_DRV_EventCaptureConfiguration()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_EventCaptureConfiguration (
    uint8 Transceiver,
    const CanTrcv_tja1145a_EventCaptureConfigType * EventConfig )
```

Writes system and CAN controller event capture configuration.

Parameters

<i>Transceiver</i>	Device ID.
<i>EventConfig</i>	Event capture configuration.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.33 TJA1145A_DRV_WakeupModeConfig()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_WakeupModeConfig (
    uint8 Transceiver,
    boolean WakeEnable )
```

Enables/disables wake up mode according to WakeEnable parameter.

Parameters

<i>Transceiver</i>	Device ID.
<i>WakeEnable</i>	Wake up functionality configuration.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.34 TJA1145A_DRV_WakeupConfiguration()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_WakeupConfiguration (
    uint8 Transceiver,
    const CanTrcv_tja1145a_WuConfigType * WakeupConfig )
```

Enables/disables WAKE pin and CAN bus wake up feature.

Parameters

<i>Transceiver</i>	Device ID.
<i>WakeupConfig</i>	Wake up configuration.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.35 TJA1145A_DRV_SetPnMode()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetPnMode (
    uint8 Transceiver,
    CanTrcv_tja1145a_PNActivationType PnMode )
```

Enables/disables partial networking functionality. If PN is activated, the device can be woken up by a remote frame. If PN is disabled, the device can be woken up only by a standard wake up pattern. It is expected that partial networking has been configured prior calling this function which only enables/disables the PN functionality.

Parameters

<i>Transceiver</i>	Device ID.
<i>PnMode</i>	Enable/disable PN functionality.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.36 TJA1145A_DRV_SetCanConfig()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetCanConfig (
    uint8 Transceiver,
    const CanTrcv_tja1145a_CanConfigType * CanConfig )
```

Configures CAN controller.

Parameters

<i>Transceiver</i>	Device ID.
<i>CanConfig</i>	Partial networking configuration.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.37 TJA1145A_DRV_SetCanControllerMode()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetCanControllerMode (
    uint8 Transceiver,
    CanTrcv_tja1145a_CanControllerModeType CanMode )
```

Sets mode of CAN controller which may differ from mode of TJA1145A. These are two entities, each with its own mode.

Parameters

<i>Transceiver</i>	Device ID.
<i>CanMode</i>	CAN controller mode.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.38 TJA1145A_DRV_ReadTimeoutFlag()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadTimeoutFlag (
    uint8 Transceiver,
    CanTrcv_tja1145a_TrsvFlagStateType * FlagState )
```

Reads timeout flag.

Parameters

<i>Transceiver</i>	Device ID.
<i>FlagState</i>	Timeout flag value.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.39 TJA1145A_DRV_ReadSilenceFlag()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadSilenceFlag (
    uint8 Transceiver,
    CanTrcv_tja1145a_TrsvFlagStateType * FlagState )
```

Reads silence flag.

Parameters

<i>Transceiver</i>	Device ID.
--------------------	------------

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.40 TJA1145A_DRV_GetTrcvSystemData()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetTrcvSystemData (
    uint8 Transceiver,
    CanTrcv_tja1145a_SysDataTypes * TrcvSysData )
```

Returns status of the transceiver (its mode, overtemperature flag) and status of CAN controller (PN status, CAN bus status, ...).

Parameters

<i>Transceiver</i>	Device ID.
<i>TrcvSysData</i>	Configuration/status data of the Transceiver.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.41 TJA1145A_DRV_ClearTrcvWufFlag()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ClearTrcvWufFlag (
    uint8 Transceiver,
    CanTrcv_tja1145a_WuReasonType WuReason )
```

Clears wake up flag. Clears one of following wake up flags (interrupts can as well wake up the device): WAKE pin flag, CAN bus wake up flag, SPI failure (this interrupt can wake up the device only when it is in standby mode), partial networking frame error, CAN bus silence.

Parameters

<i>Transceiver</i>	Device ID.
<i>TrcvSysData</i>	Configuration/status data of the transceiver.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.42 TJA1145A_DRV_GetPorEventStatus()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetPorEventStatus (
    uint8 Transceiver,
    CanTrcv_tja1145a_PorEventType * PorEventStatus )
```

Reads power on event flag.

Parameters

<i>Transceiver</i>	Device ID.
<i>PorEventStatus</i>	POR event status.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.43 TJA1145A_DRV_ClearPorFlag()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ClearPorFlag (
    uint8 Transceiver )
```

Clears power on event flag.

Parameters

<i>Transceiver</i>	Device ID.
--------------------	------------

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.44 TJA1145A_DRV_GetSystemErrorStatus()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_GetSystemErrorStatus (
    uint8 Transceiver,
    uint16 * SystemError )
```

Returns system error status. This driver recognizes following system errors:

- overtemperature, which forces the device to switch to overtemperature mode,
- undervoltage on VCC and/or VIO, which forces the device to shift to sleep mode,
- oscillator of CAN transceiver not running at targeted frequency,
- VCC undervoltage detected by CAN transceiver.

CAN failure

- CAN bus silence
- partial networking error
- partial networking configuration error

Parameters

<i>Transceiver</i>	Device ID.
<i>SystemError</i>	Stores information about system error.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.45 TJA1145A_DRV_ClearSystemErrorStatus()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ClearSystemErrorStatus (
    uint8 Transceiver,
    CanTrcv_tja1145a_SystemErrorType SystemError )
```

Clears system error. Important: it is not possible to clear all system errors. Following system errors are read only flags: CAN oscillator status, VCC undervoltage, PN configuration error (in this case PN must be reconfigured). Note: TJA1145A_PN_ERROR flag can be set because of two reason:

- PN frame error was detected - the flag can be cleared
- PN configuration error was detected - PN must be reconfigured to clear the flag.

Parameters

<i>Transceiver</i>	Device ID.
<i>SystemError</i>	System error to clean.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.46 TJA1145A_DRV_SetLockControlRegister()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_SetLockControlRegister (
    uint8 Transceiver,
    const CanTrcv_tja1145a_LockRegistersType * LockControl )
```

Writes lock configuration to prevent unintended register access.

Parameters

<i>Transceiver</i>	Device ID.
<i>LockControl</i>	Lock control configuration.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.47 TJA1145A_DRV_WriteMemoryRegister()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_WriteMemoryRegister (
    uint8 Transceiver,
    CanTrcv_tja1145a_MemoryRegisterType MemRegister,
    uint8 Data )
```

Writes general purpose memory register.

Parameters

<i>Transceiver</i>	Device ID.
<i>MemRegister</i>	Which memory register to access.
<i>Data</i>	Data to write to given memory register.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.48 TJA1145A_DRV_ReadMemoryRegister()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_ReadMemoryRegister (
    uint8 Transceiver,
    CanTrcv_tja1145a_MemoryRegisterType MemRegister,
    uint8 * Data )
```

Reads general purpose memory register.

Parameters

<i>Transceiver</i>	Device ID.
<i>MemRegister</i>	Which memory register to access.
<i>Data</i>	Data to store read out content of a memory register.

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.5.49 TJA1145A_DRV_Deinit()

```
CanTrcv_tja1145a_StatusType TJA1145A_DRV_Deinit (
    uint8 Transceiver )
```

Deinitializes the TJA1145A driver. SID is 0x0F. Clears the internal run-time configuration and puts the device to standby mode.

Parameters

<i>Transceiver</i>	Device ID.
--------------------	------------

Returns

CanTrcv_tja1145a_StatusType Return code.

6.1.6 Variable Documentation

6.1.6.1 DeviceNum

```
uint8 DeviceNum
```

Number of devices controlled by the driver. The data type is uint8, because Autosar API indexes connected devices with an index of type uint8.

Definition at line 211 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.2 ConfigData

```
const CanTrcv_tja1145a_ConfigurationDataType* ConfigData
```

Pointer to an array of device and SPI configuration structures.

Definition at line 213 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.3 WakeupSourceEn

`boolean WakeupSourceEn`

Wake up source enabled/disabled.

Definition at line 224 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.4 EcuMWakeupSource

`EcuM_WakeupSourceType EcuMWakeupSource`

Reference to the wake up source defined in the ECU state manager.

Definition at line 226 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.5 WakeupSourceConfig

`CanTrcv_43_tja1145a_WakeupSourceRefType WakeupSourceConfig`

Configuration of standard wake up source used in reports to ECU manager.

Definition at line 237 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.6 PorWakeupSourceConfig

`CanTrcv_43_tja1145a_WakeupSourceRefType PorWakeupSourceConfig`

Configuration of POR wake up source used in reports to ECU manager.

Definition at line 239 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.7 SysErrWakeupSourceConfig

`CanTrcv_43_tja1145a_WakeupSourceRefType SysErrWakeupSourceConfig`

Configuration of SYSERR wake up source used in reports to ECU manager.

Definition at line 241 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.8 DemErrorReportConfig

Mcal_DemErrorType DemErrorReportConfig

DEM reporting configuration.

Definition at line 244 of file [CanTrcv_43_tja1145a.h](#).

6.1.6.9 CanTrcv_43_tja1145a_eDriverStatus

volatile [CanTrcv_43_tja1145a_eDriverStatusType](#) CanTrcv_43_tja1145a_eDriverStatus [extern]

Holds current status/run-time of CANTRCV driver: CANTRCV_43_TJA1145A_UNINIT or CANTRCV_43_↔TJA1145A_READY.

6.2 Spi_external_access

6.2.1 Detailed Description

Function Reference

- Std_ReturnType [CanTrcv_Ip_SpiTransferExternal](#) (const CanTrcv_Ip_SpiSetupType *pSpiSetup, const uint8 *pTxBuffer, uint8 *pRxBuffer)
SPI transfer.
- Std_ReturnType [CanTrcv_Ip_SpiSetAsyncMode](#) (void)
Sets the SPI asynchronous mode.
- void [CanTrcv_Ip_SpiNotifyTransferEnd](#) (void)
Notifies the SPI transfer end.

6.2.2 Function Reference

6.2.2.1 CanTrcv_Ip_SpiTransferExternal()

```
Std_ReturnType CanTrcv_Ip_SpiTransferExternal (
    const CanTrcv_Ip_SpiSetupType * pSpiSetup,
    const uint8 * pTxBuffer,
    uint8 * pRxBuffer )
```

SPI transfer.

This function transfers data via SPI.

Parameters

in	<i>pSpiSetup</i>	SPI interface setup
in	<i>pTxBuffer</i>	SPI Tx buffer
out	<i>pRxBuffer</i>	SPI Rx buffer

6.2.2.2 CanTrcv_Ip_SpiSetAsyncMode()

```
Std_ReturnType CanTrcv_Ip_SpiSetAsyncMode (  
    void )
```

Sets the SPI asynchronous mode.

This function sets the SPI asynchronous mode.

6.2.2.3 CanTrcv_Ip_SpiNotifyTransferEnd()

```
void CanTrcv_Ip_SpiNotifyTransferEnd (  
    void )
```

Notifies the SPI transfer end.

This function notifies the SPI transfer end.

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